

# High-Resolution Laminar Identification in Macaque Primary Visual Cortex Using Neuropixels Probes

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## eLife Assessment

This **important** study provides insights and strategies for assessing laminar structure in vivo in the visual cortex of the macaque monkey with high-density linear electrode arrays. The paper provides **convincing** evidence demonstrating that signals in higher frequency bands, related to the discharge of action potentials, are of substantially better use for achieving well-resolved cortical layer identification than are signals in lower frequency bands typically associated with local field potentials and standard-practice Current Source Density (CSD) analyses. These findings are of interest to a wide range of neuroscientists making comparisons between cortical layers or recording with array electrodes.

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## Abstract

Laminar electrode arrays allow simultaneous recording of activity of many cortical neurons and assignment to layers using current source density (CSD) analyses. Electrode arrays with 100-micron contact spacing have been used to estimate borders between layer 4 versus superficial or deep layers, but in macaque primary visual cortex (V1) there are far more layers, such as 4A which is only 50-100 microns thick. Neuropixels electrode arrays have 20-micron spacing, and thus could potentially discern thinner layers and more precisely identify laminar borders. Here we show that laminar distributions of CSDs lack consistency and the spatial resolution required for thin layers and accurate layer boundaries. To take full advantage of high density Neuropixels arrays, we have developed approaches based on higher resolution electrical signals and analyses, including spike waveforms and spatial spread, unit density, high-frequency action potential (AP) power spectrum, temporal power

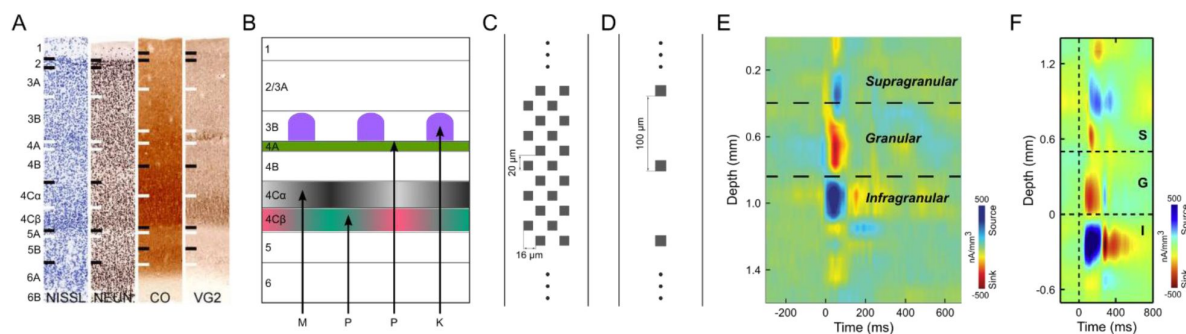
change, and coherence spectrum, that afford far higher resolution of laminar distinctions, including the ability to precisely detect the borders of even the thinnest layers of V1.

## Main Text

The laminar organization of the neocortex reflects the distributions of cells with different soma shapes and sizes, packing densities, and patterns of synaptic inputs and axonal outputs (1, 2) (**Figure 1A**). To link functional properties to cell types and circuits it is useful to identify the laminar locations of recorded cells. Historically this was first done by using a single electrode and marking recording locations with periodic lesions that are typically rather large and asymmetric, leaving considerable uncertainty. In the context of visual cortical circuits and function, the visual receptive fields of neurons have often been characterized with respect to laminar depth (3–5).

The development of laminar electrode arrays has allowed simultaneous recordings from many locations across cortex enabling current source density (CSD) analysis (6, 7) based on Local Field Potentials (LFP) in response to full-field flashed visual stimuli. The estimated laminar boundaries from CSD profiles allow direct assignment of units recorded on particular electrode contacts to their cortical layers (8–10). Typically, a fast current sink in the CSD profile following a full-field dark to light visual stimulus transition is attributed to the lateral geniculate nucleus (LGN) thalamic-recipient granular layer (L4) and the top and bottom of this sink are used to specify supragranular and infragranular divisions. The recent advent of long and very high-density Neuropixels arrays (11) has the theoretical potential to allow the identification of more layers across greater cortical depths and higher resolution identification of laminar boundaries. This potential could be particularly impactful for laminar identification in the primary visual cortex (V1) of monkeys where V1 is composed of multiple thin and functionally disparate sublayers, including at least 5 distinct layers receiving input from functionally different LGN afferent types (12, 13). Neurons in L4C $\beta$  of V1 have small cell bodies and receive input from LGN parvocellular L/M cone-opponent afferents that terminate in 4C $\beta$ , whereas neurons in 4C $\alpha$  have slightly larger cell bodies and receive LGN magnocellular achromatic inputs that terminate only in 4C $\alpha$ . The thin layer 4A (L4A) is packed with small granule cells that receive parvocellular S-cone OFF (S-OFF) inputs, and layer 3B (L3B) cells in high Cytochrome Oxidase (CO) intensity blobs receive LGN koniocellular S-cone ON (S-ON) projections (**Figure 1B**). Layer 6 (L6) also receives direct LGN input and can be divided into upper (L6A) and lower (L6B) halves based on the higher density of cell bodies in 6A (12, 14).

Previous use of laminar electrode arrays and CSD analyses for laminar assignment in monkey V1 (8–10) is limited (**Figure 1E, F**), because the vertical spacing of electrodes is too large (e.g. 100 $\mu$ m) to precisely assign laminar boundaries or to detect thin layers such as L4A, which is less than 100  $\mu$ m thick. Moreover, with sparse electrode contacts, only a few single units are isolated from layers such as 4C $\beta$  of V1, which are composed of very small neurons (~5–8 $\mu$ m in diameter). In this study, Neuropixels probes (11), with 384 simultaneously active recording channels and 20 $\mu$ m vertical spacing spanning 3.84mm were used to record representative samples of neurons across the full depth of V1 (**Figure 1C**). A prior study used CSD analysis with Neuropixels electrodes for laminar identification in V1 of macaque monkeys (15), but did not attempt to take advantage of the potential for high-resolution laminar assignment; instead, recordings were averaged across four contacts, smoothed ( $\sigma$ =120 $\mu$ m), and laminar boundaries approximated based on expected depth from a single CSD transition attributed to the granular/infragranular border. It might be expected that high-resolution CSD profiles recorded using Neuropixels with 20 $\mu$ m vertical electrode spacing can provide finer details to identify more V1 layers with higher accuracy. In the present study, however, we show that CSD profiles are low resolution and



**Figure 1.**

### Cortical layers of V1.

(A) Brodmann's laminar divisions of macaque monkey V1 revealed with different histological staining methods (Nissl for cell density, Anti-NeuN for neuronal density, CO for metabolic activity and VG2 for glutamatergic axon terminals from thalamus). Adapted from (1). (B) Schematic diagram of V1 layers that receive different functional/anatomical types of LGN afferent input (13). LGN magnocellular (*M*; *L+M achromatic*, grey in 4Ca), parvocellular (*P*; *L/M cone-opponent*, red-green in layer 4Cb; *S-OFF*, lime in layer 4A) and koniocellular (*K*; *S-ON*, purple in layer 3B CO blobs). (C) Electrode spacing and layout of Neuropixels 1.0 probe. (D) Layout of probes that were used in E and F. (E) An example of V1 CSD profile in awake recordings using a 16-channel linear probe (8). (F) Another example of V1 CSD profile in anesthetized recordings using a 24-channel linear probe (9). Supragranular (S), granular (G) and infragranular (I) layers were delineated in E and F mainly based on transitions between the short-latency granular layer sink and adjacent sources.

inconsistent across both penetrations and visual stimulus conditions, making it impossible to summarize a stereotypical CSD profile that can be used to reliably and accurately relate electrode contacts to distinct layers and sublayers in V1.

In physiological situations, cortical LFPs arise mainly from synchronized synaptic activities (16). Because synapses are widely distributed on dendrites, the current sinks and current sources of synaptic events revealed by CSD can be far from the cell body (17). By convention, anatomical properties, such as neuron density (18) and soma size/shape are the main metrics used for anatomical laminar structure identification. Therefore, the boundaries between sinks and sources of CSD profiles do not necessarily correspond closely to laminar borders using anatomical features. Here we describe the use of alternatives to CSD profiles for linking electrical activity to layers, with the key point being the use of signals with higher-spatial resolution than CSDs and potentially having a better match to layers identified with cell body sizes, and cell densities in histological staining. High frequency (300-3000Hz) Action Potential (AP) band signals are related to the combined voltage fluctuations resulting from neuronal action potential firing, with the largest amplitude signals originating at cell bodies. These high-frequency signals have much faster attenuation across space compared to LFPs (17), potentially allowing for higher spatial resolution. Thus, we formulated several metrics based on AP band signals and found that they show various degrees of similarity to the laminar distributions of neuron density and soma size/shape across cortical layers. Since some layers share similar anatomical properties and are hard to differentiate even in histological staining, we also deployed colored visual stimuli that preferentially activate LGN afferents terminating in specific layers (Figure 1B). By combining selected identification metrics under different stimulus conditions, we show how high-resolution laminar structures of V1, including major layers and sublayers, can be consistently and accurately identified *in vivo* without the assistance of postmortem histological staining.

## Results

35 vertical electrode penetrations (at least 1mm apart, typically far more, see Figure S11) were made using Neuropixels 1.0 probes in primary visual cortex (V1) of two anesthetized/paralyzed rhesus macaque monkeys (17 penetrations in monkey A and 18 in monkey B), of which 31 are included in our analyses (See Methods for criteria used to exclude penetrations A7, A8, A17 and B18). Of these 31 penetrations, 20 were centered on Cone Opponent Functional Domains (COFD) and 9 on achromatic ON/OFF domains revealed by intrinsic signal imaging (ISI) (19). The remaining 2 penetrations were targeted to regions outside these domains. ISI was also used to reveal ocular dominance columns in V1 and the border between primary and secondary (V2) visual cortex. We will first present results using CSD analysis demonstrating its limitations in spatial resolution and consistency, and then introduce new metrics based on AP band signals to effectively distinguish between even thin V1 layers and to precisely define laminar boundaries. We then describe a set of steps for applying these metrics to guide identification of all laminar boundaries in V1. Note that the labeled layer boundaries in all figures, including those illustrating CSD data, were determined using the full set of metrics; therefore these plots illustrate borders that might not be apparent solely from visualization of the analyses being illustrated in a particular plot. Discrepancies between labeled layer boundaries and borders revealed in plots based on particular metrics highlight the limitations of those metrics, as is most clear from the CSD analyses described below.

### CSD profiles are low-resolution and inconsistent for the same stimulus

The shortest latency current sink in cortical area V1 CSD profiles evoked from flashed, full-field visual stimuli is thought to correspond roughly to L4C, which receives the most dense LGN inputs; and the transition from this sink to a deeper current source has been used to identify the border

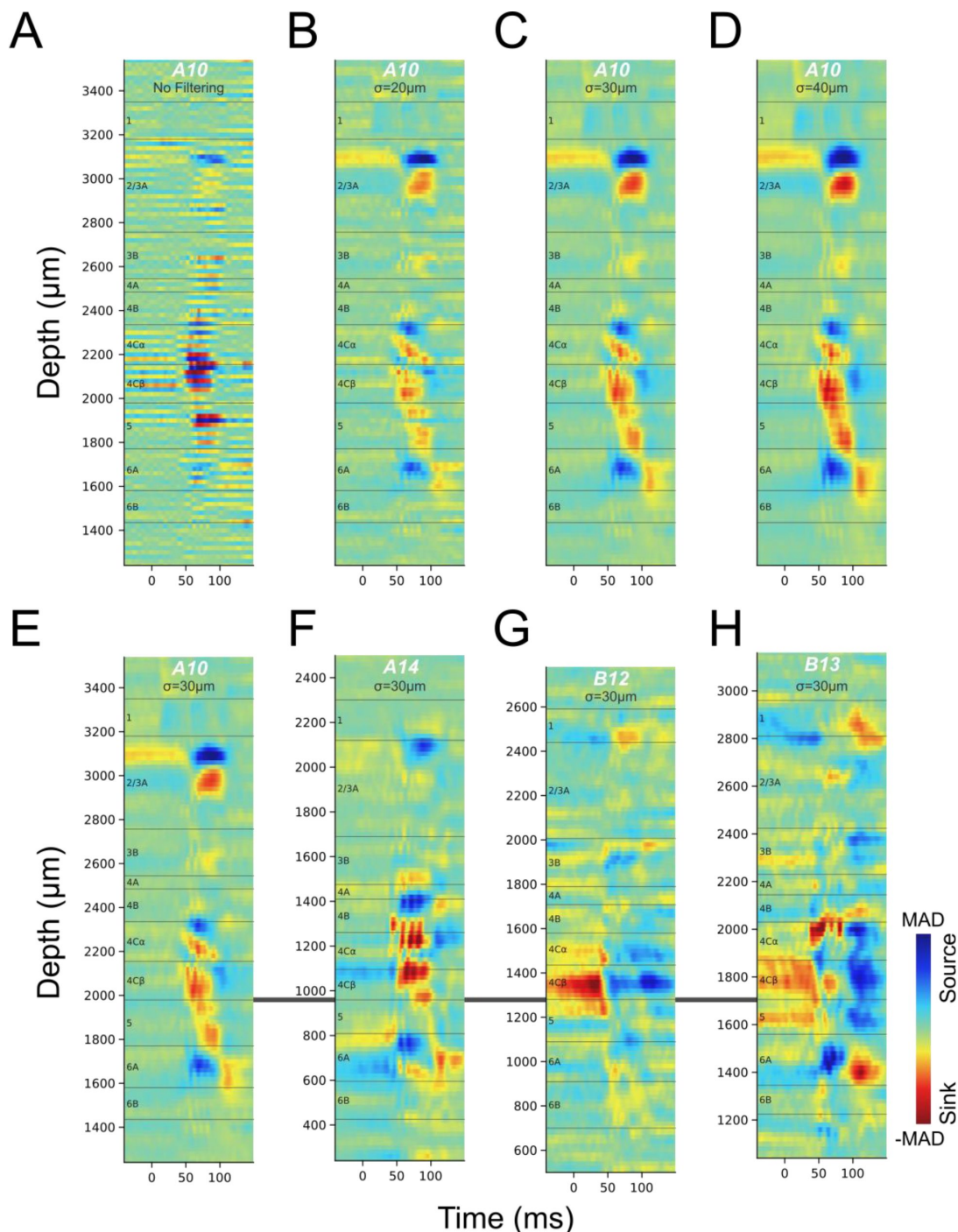
between granular L4 and infragranular L5 (8–10, 15). CSD profiles (Figure 1E, F) in many previous studies were obtained using 16, 24, or 32-channel linear probes with 100µm vertical spacing (Figure 1D). The signal detection gap in these electrodes precludes identification of thin layers or sub-layers such as 4A, 4Ca and 4Cβ. Interpolations were usually applied to these CSD profiles so that layer borders could be drawn at sink-source boundaries at a resolution higher than 100µm, however, the location of a sink-source boundary might deviate substantially from the same boundary in CSD profiles produced by Neuropixels, which have a 20µm vertical spacing. Figure 2A shows an example 20µm spacing CSD profile obtained using Neuropixels. At this resolution the CSD signals are highly fragmented due to more or less locally balanced high resolution sink-source patterns along the probe. While there appear to be alternating contacts with a strong short latency sink within L4C, these only span a depth of about 200µm, far narrower than L4C, leaving considerable uncertainty about where boundaries should be drawn. To simulate results with lower resolution arrays and to potentially facilitate layer identification by eliminating local current spread, we applied gradually increasing width ( $\sigma=20, 30, 40\mu\text{m}$ ) gaussian filters to derive sink-source patterns that more closely matched the scale of most cortical layers. In figures 2B–D, it can be seen that, overall, CSD profiles become more stable with increasing degrees of smoothing while missing small details. This gaussian filtering approach yielded nearly equivalent results to decreasing spatial resolution by calculating CSDs using down-sampling to skip 1, 2, 3, or 4 contacts, which mimicked the CSDs that would have been generated by linear probes with 40, 60, 80, or 100µm spacing (Figure S1). To better match the scale of cortical layers while keeping as much detail as possible, all subsequent CSD analysis results illustrated were gaussian filtered with  $\sigma=30\mu\text{m}$ , which was almost identical to CSD results from analyses with down-sampling at ~60µm contact spacing.

The most used visual stimulus in past studies is a full screen switching from black to white. We therefore first compare our spatially smoothed CSD profiles using black to white transitions with the low spatial resolution CSD profiles (Figure 1E, F) reported in previous studies (8, 9) in which a stimulus presentation screen was also flipped from black to white for ~100ms. Figures 2E–H show CSD profiles from four penetrations generated by flashing from a black to a white screen through the dominant eye (black to white every 3 seconds); these CSDs clearly lack consistency across penetrations. The criteria from previous studies using the best boundary between an initial sink and a deeper source to mark the border between L4C and L5 is not tenable because sometimes no such transition is apparent and there is never a clearly identifiable transition. More importantly, the observed transitions do not consistently match to layer identification using the methods that we establish and validate below. For example, in Figure 2E, criteria applied in previous studies would mark the bottom of L4C at 1800µm, where the initial sink transitions to a source (compare Figure 1E to 2E). But this border location is highly unlikely because there would be insufficient space to accommodate the known thickness of L5 and L6 before our unambiguous identification of the L6/white matter border (see below). Furthermore, in some cases the CSD profile in response to a black to white screen has a zero latency sink in L4C that transitions to a current source (Figures 2G, H). A prior study has shown that the zero latency sink corresponds to residual signal from the prior white to black transition (20) which occurs 1.5 seconds earlier in our experiments. We note that this signal is much stronger in layer 4Cβ than 4Ca, consistent with more sustained activity in parvocellular LGN neurons and an asymmetry in which there is more sustained activity in the dark (black stimulus) condition.

## Inconsistency and stimulus dependency of CSD profiles

Here we explore the consistency of CSD profiles both across individual penetrations and across visual stimulus conditions. In view of the ambiguity in CSD profiles that might persist beyond the time before a new stimulus transition (Figure 2G, H), we use  $\Delta\text{CSD}$  (See Methods) in subsequent CSD analyses to allow comparisons of CSDs under additional stimulus conditions and to further assess variability in CSD responses. In addition to conventional black and white stimuli, we constructed L/M cone-opponent colors and S/(L+M) cone-opponent colors, to preferentially activate





**Figure 2.**

**CSDs evoked by a black to white visual stimulus transition through the dominant eye, all targeting achromatic ON domains.**

The layer boundaries drawn here were determined using the methods and data described in the subsequent sections. (A) CSD profile from the LFP traces of all contacts with 20μm depth spacing (Penetration A10). (B-D) Gaussian filtering of the CSD profile in A along the depth of probe with  $\sigma$  equals 20μm (B), 30μm (C) or 40μm (D). (E-H) CSD profiles from four penetrations (E: A10, F: A14, G: B12, H: B13) generated using 30μm gaussian smoothing. The CSD profiles were aligned relative to the border between L4Cβ and L5, and CSD scales were each independently color-mapped relative to the maximum absolute deviation (MAD) in each profile.

cone-opponent parvocellular LGN inputs to L4C $\beta$  (L/M opponent) or L4A (S-OFF), or koniocellular LGN inputs (S-ON) to L3B blobs (**Figure 1B** [↗](#)). **Figure 3** [↗](#) shows  $\Delta$ CSD profiles generated in response to various stimulus conditions for three penetrations. Within each penetration, different stimulus conditions usually evoked different  $\Delta$ CSD profiles, some of which could differ drastically, e.g. black vs. white (**Figure 3C** [↗](#)), or dominant eye vs. non-dominant eye (**Figure 3B** [↗](#)). Meanwhile,  $\Delta$ CSD profiles generated using the same color or ocular dominance also show various degrees of inconsistency across penetrations. The most consistent and clear generation of a fast sink corresponding to L4C is observed with a black stimulus (white to black) presented to the dominant eye, whereas the response to a white stimulus is more variable and can evoke an early L4C source in some penetrations (**Figure S2, S3** [↗](#)). Note that even in these best-case conditions, the L4C sinks leave considerable uncertainty about where laminar boundaries should be drawn.

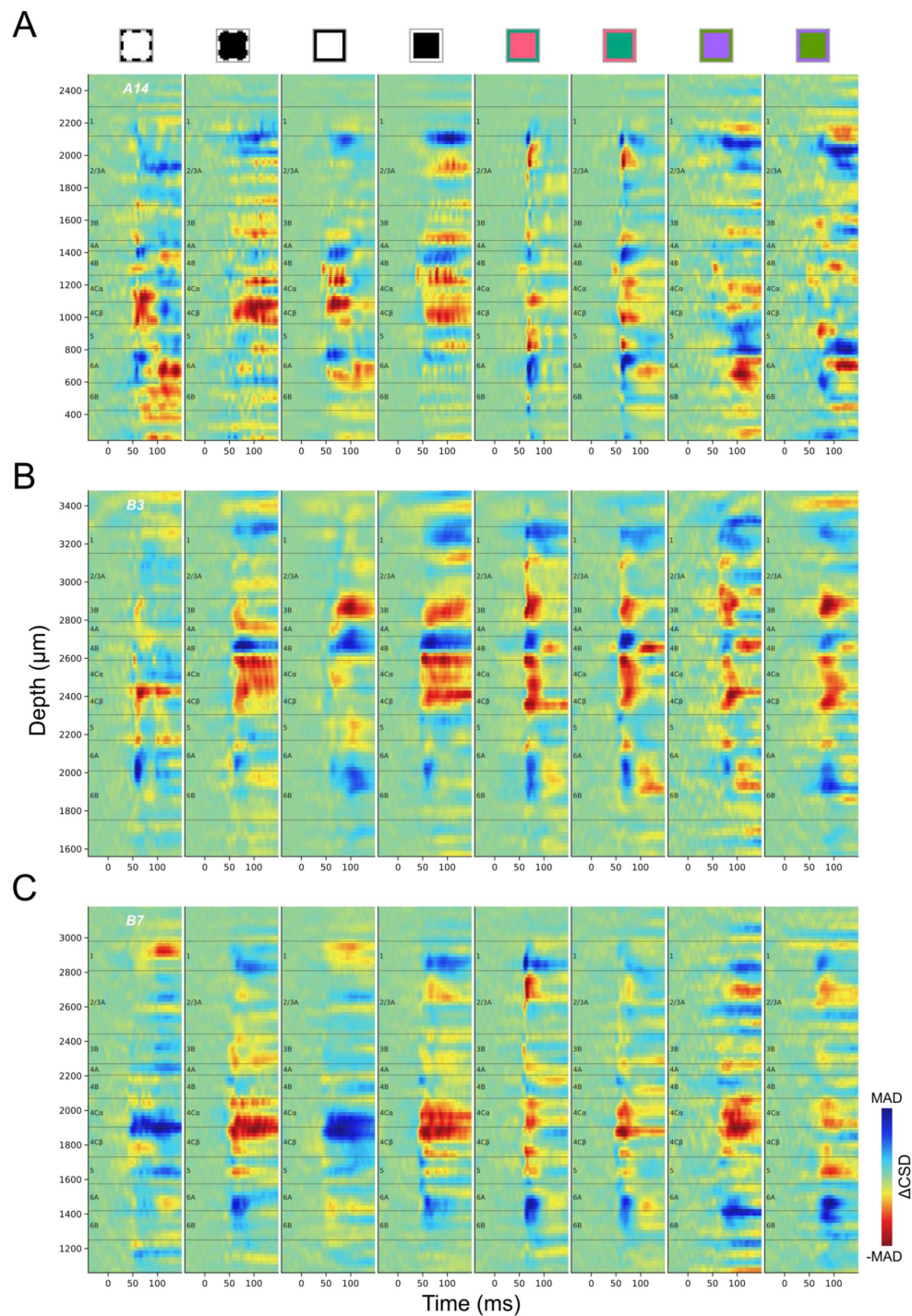
Responses to L/M cone-opponent colors also differ from black/white responses in that they often lack a current source below L4C that could help to identify the L4C/L5 border (**Figure 3** [↗](#)). The variability of  $\Delta$ CSD profiles across penetrations makes it difficult to summarize a single stereotypical profile that can be used as a template to delineate layer boundaries for each penetration. Even using the most stable stimulus, black screen to the dominant eye, landmarks are often missing that would be needed to reliably identify major layers and sublayers. These results suggest that new methods are required to accurately and consistently identify most of the known layers and sublayers.

## Action potential-based metrics for laminar identification

In the following sections we describe several metrics extracted from high frequency action potential (AP) band signals that can be informative for more definitively identifying the precise location of laminar borders and for identifying borders that are not typically seen with conventional CSD analyses. These metrics can be categorized into two groups. Metrics in the first group were features derived from the density and spike shapes (“waveforms”) of units identified through spike sorting. These were independent of the visual stimuli used and reflected the inherent properties of spikes and the spatial relationships between locations of spike initiation/propagation and electrode contacts. Metrics in the second group were derived directly from AP signal responses to full-field flashed visual stimuli. We also compared these measures for responses to different visual stimulus conditions designed to potentially reveal locations receiving input from different types of LGN afferents.

## Unit Density

The neuron packing density is a clear anatomical property that is useful for differentiating cortical layers with histological staining. Neuron density depth distributions derived by counting immunolabeled neurons in confocal microscopy (18 [↗](#)) clearly show packing density patterns that corresponded to each layer. In macaque monkey V1, L4C $\beta$  has the highest packing density. Moving downward, there is a density trough in L5 and then an increase again in L6. Moving upward from L4C $\beta$ , density decreases across L4C $\alpha$  and L4B, until it rises to a small peak that corresponds to the thin L4A. To estimate neuron density from our electrical recordings we can plot the density of units resolved during spike sorting. The accuracy of this estimation depends, of course, on the ability to consistently resolve large numbers of units across the depth of cortex and is also subject to potential sampling biases that might make some spike waveforms easier to isolate than others. **Figure 4A** [↗](#) plots unit density (See Methods) for an example penetration across V1 and into the white matter (WM), and shows clear changes with depth that partially capture the shape of the expected neuron density distribution and provide landmarks helpful for inferring the likely identities of some layers. The obvious peak around the electrode depth at 2400 $\mu$ m (distance relative to deepest active electrode contact) in **Figure 4A** [↗](#) corresponds to L4C and the trough below that peak is expected from the low neuron density in L5. Below L5, the expected transition to higher density in L6A is observed, as well as decreased density in L6B. These same transitions can be observed in the histological profiles of neuron density described in **figure 6** [↗](#) of the study



**Figure 3.**

**$\Delta$ CSD profiles of three penetrations (A: A14, B: B3, C: B7) under different stimulus presentations.**

Square icons above CSD profiles indicate the different color transitions used to generate the CSDs, as well as which eye was stimulated. The outline of each square indicates the color of the screen before the stimulus transition at time zero, while the filled color indicates the flashed color (Achromatic, Black and White; or color pairs in DKL color space; See Method). Black and White correspond to achromatic stimuli; Red to L-ON, M-OFF; Green to M-ON, L-OFF; Purple to S-ON, (L+M)-OFF; and Lime to S-OFF, (L+M)-ON. Dashed contours (two leftmost columns) indicate visual presentation to the non-dominant eye and solid contours indicate stimuli presented to the dominant eye. Color maps for  $\Delta$ CSD profiles were independently scaled according to the maximum absolute deviation (MAD) in each profile.



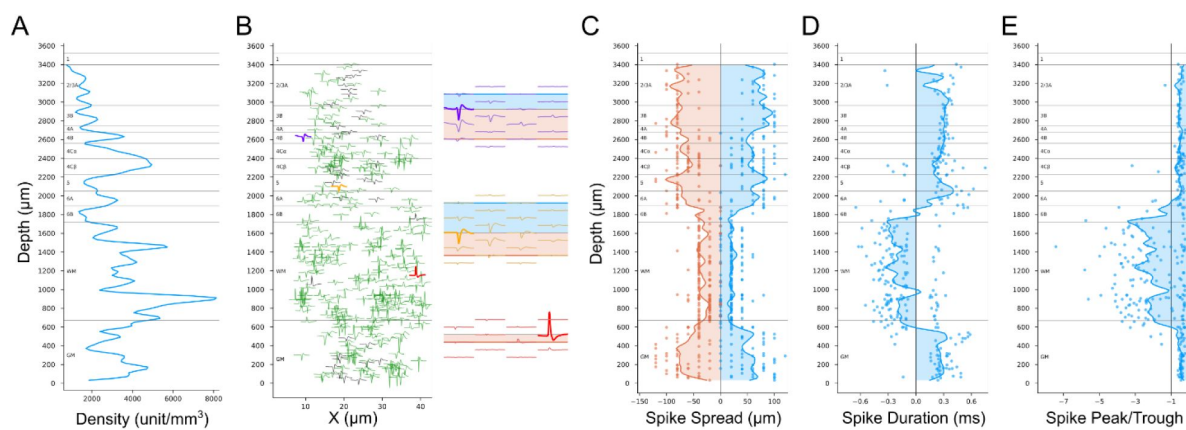
from Kelly and Hawken (18 [↗](#)); note however that the electrical signals contrast strongly with the histological density at the L6/WM border because high densities of units isolated from axons can be detected in the white matter. As shown below, these can be distinguished by their unique spike waveforms.

## Metrics from Spike Waveforms

Cell body size and shape are additional anatomical features related to laminar changes in histological staining. Densely packed neurons usually have small cell bodies, e.g., the majority of neurons in L4C $\beta$  and L4A are small granule neurons. Conversely, sparsely packed neurons in L4B and L5 are mostly large pyramidal or spiny stellate neurons. The spiking signal of a neuron propagates through extracellular space with decaying amplitude. The travel distance before it is attenuated to the noise level can be measured in a group of electrodes of which the spacing is less than the decay distance (**Figure 4B** [↗](#) insets). The spike spatial spread measured on electrodes may be related to soma size/shape. Assuming two neurons are located at the same horizontal distance from the probe, then the one with the larger soma, or a more elongated shape along the axis of the probe should be detected on more electrode contacts, resulting in larger spike spread.

**Figure 4C** [↗](#) plots the distributions of spike spread distances measured for units isolated on contacts at different laminar depths. Although the distributions are somewhat granular owing to the fixed 20 $\mu$ m spacing between electrode contacts, the smoothed mean distribution is clearly at its lowest in the white matter, and within the cortex the lowest mean values are observed across L4C. There is also an increase in spread from L4C to L5 as expected from the spikes from larger L5 cells. While other transitions from this measure do not always have a close match to the boundaries identified with other measures, spike spread does prove to be particularly useful to differentiate white matter from gray matter because the spread of axonal spikes or noise detected in white matter is much smaller than that of spikes from cell bodies. The location of the L6/WM border is further corroborated by the expected transition to inverted spikes that are measured from axons in the white matter (**Figure 4B** [↗](#), compare red inverted spikes to conventional purple and orange spikes). These inverted spikes have a negative spike duration (difference in time from peak to trough) and a large, negative peak-trough ratio ( $< -1$ ). Depth plots of spike duration (**Figure 4D** [↗](#)) and peak-trough ratio (**Figure 4E** [↗](#)) show clear and sharp transitions at the L6/WM border. In these examples, there is an additional transition as the electrode penetrates across the white matter and back into L6 of the V2 cortex that is folded under the opercular cortical surface. Note that a few inverted spikes are also detected in a L6B transition zone where cell bodies are intermingled with a larger density of axons than in more superficial layers, as well as in L4C $\beta$  where parvocellular LGN afferents are very dense and their spikes can be isolated relative to the surrounding small spikes from 4C $\beta$  spiny stellate neurons.

Unit density, spike spread, spike duration and spike peak-trough ratio are all stimulus-independent spike features, and together can be used to identify the approximate locations of L4C and L5, as well as the L6/WM border. These are crucial landmarks that can serve as anchors for the validation and assignment of the identities of additional borders and layers derived from other features. Use of this group of metrics for laminar assignment depends on numbers of detected and resolved units; thus, probes with high contact numbers and densities, such as Neuropixels, are essential. In practice, cortical health is also important as damage or conditions reducing the number of active and resolvable spikes will drastically reduce the number of detectable units, and the spike sorting process may exclude small amplitude spikes or unresolvable spikes further reducing the yield, in turn impairing the accuracy of estimating neuron density and soma size/shape through these metrics. We next explore a second group of complementary metrics derived directly from stimulus-evoked AP band signals that may resolve additional laminar boundaries and/or display sharper transitions at laminar borders. These measures also circumvent limitations imposed by numbers of units detected and resolved and might prove useful with lower density electrode arrays.



**Figure 4.**

### Unity density and spike waveform features across cortical depth.

(A) Unit density distribution along the probe for one penetration (A6). (B) Average spike waveform of maximum height was shown for single unit (green) or multi-unit (gray). The insets show the spike spatial spread on the Neuropixels probe for the units with corresponding color, and the filled bands indicate the upward spread (blue) and downward spread (orange). (C) Upward (blue) and downward (orange) spike spread. (D) Spike duration. (E) Spike peak-trough amplitude ratios. A-E are from the same penetration (A6). In C-E, each dot represents a unit, and lines are smoothed distributions along the probe (See Methods). The transitions to small spike spread in C, sign reversals of spike duration in D and crosses at  $-1$  of peak-trough ratio in E, are strong indicators for the locations of the border between white matter (WM) and gray matter (GM).

## Power Spectrum of AP Signals

Below we introduce three additional AP-band metrics that are all derived from the high-frequency voltage fluctuations measured in the AP band (300-3000Hz). Two of these metrics, “AP Power” and “Coherence”, each generate spectral plots based on voltages measured across cortical depth, plotted against the frequencies of the voltage fluctuations. Because these plots show the average values during a response window (10-150ms after the visual stimulus transition), they are insensitive to temporal differences in responses that might be expected due to different response latencies at different depths, such as faster transitions expected at LGN recipient layers or layers receiving shorter latency LGN inputs from the faster propagating magnocellular afferents that terminate in L4Ca.

In cortical tissue, AP band voltage fluctuations result mainly from the superposition of neuronal spikes, which typically feature a trough-peak deflection of extracellular potentials (**Figure 4B** [↗](#) purple and orange). There are several properties of spiking activity evoked by visual stimuli that are expected to influence laminar power spectrum profiles. First, there is expected to be an overall influence of neuron density. For example, if every spike were to have the same shape and amplitude and there were no waveform cancelling due to temporal overlap, then the Root Mean Square (RMS) amplitude of the AP signal on a recording channel would be directly proportional to number of spikes. This factor might be expected to result in some similarity between laminar power profile and that of neuronal density, given the firing rates of neurons across layers are also similar. Second, the number of spikes detected is also expected to reflect the degree to which neurons close to each contact are activated by the particular visual stimulus that preceded the response window. Since all visual stimuli used were full-field and not spatially structured, this might be expected to result in a strong relationship to layers that respond to full-field stimuli, as well as those receiving direct input from LGN axons responding to these stimuli and whose APs also contribute to measured voltage changes. Taken together, it is possible that the firing rate profile across depth could counteract the depth profile of neuron density, and the resulting depth profile of power could deviate far from the depth profile of neuron density. Nevertheless, we found that depth profile of firing rate to these full-field stimuli roughly matches the depth profile of neuron density (**Figure S4B, C** [↗](#)), making power spectrum profile a useful estimator of neuron density.

**Figure 5** [↗](#) plots RMS<sup>2</sup> Power Spectra ( $P_f$ , See Methods) during a response window from 10-150ms after various stimuli, averaged across trials. In the majority of penetrations, laminar power spectra plots have at least two clear laminar bands corresponding to L4C and L6A. Under some stimulus conditions an additional thin band is seen in L4A. These signatures almost always precisely align with sharp transitions at the top and bottom of L5, marking the L4Cβ/L5 border and the L5/L6 border. In most cases the top of the L4C band corresponds to the L4B/L4C border. But in some cases (e.g. **Figure 5B** [↗](#)) the transition to lower power occurs below the expected L4B/L4C border as identified from other measures from the same penetration, such as response latencies and the known thickness of L4Ca relative to L4Cβ (see below). In cases where a thin L4A band is visible, the borders can precisely identify the top and bottom of that layer. These patterns match with the expectation of lower AP power in layers with low neuronal density that do not receive direct LGN input (e.g. L4B and L5). Insofar as both L6A and L6B receive direct LGN inputs ([14](#) [↗](#)), the greater AP power in 6A is likely due to higher neuron density. The stronger signal that is sometimes observed in L4Cβ relative to L4Ca might also be expected from the higher neuronal density in L4Cβ, but this is not a consistent feature of laminar power spectra. It is also possible that differential patterns under various stimulus conditions might relate to the columnar organization or laminar organization of cone-opponent mechanisms in V1. For example, S-OFF LGN inputs to L4A might be expected to result in consistent L4A activation to the S-OFF stimulus, but we do not find highly predictable relationships between the particular-colored stimuli that differentially

activate layers to the ISI-identified cone-opponent columns targeted by the electrode penetrations. Nevertheless, the use of multiple stimuli appears to increase the chances of detecting these unpredictable activation patterns.

## Local Coherence spectrum of AP Signals

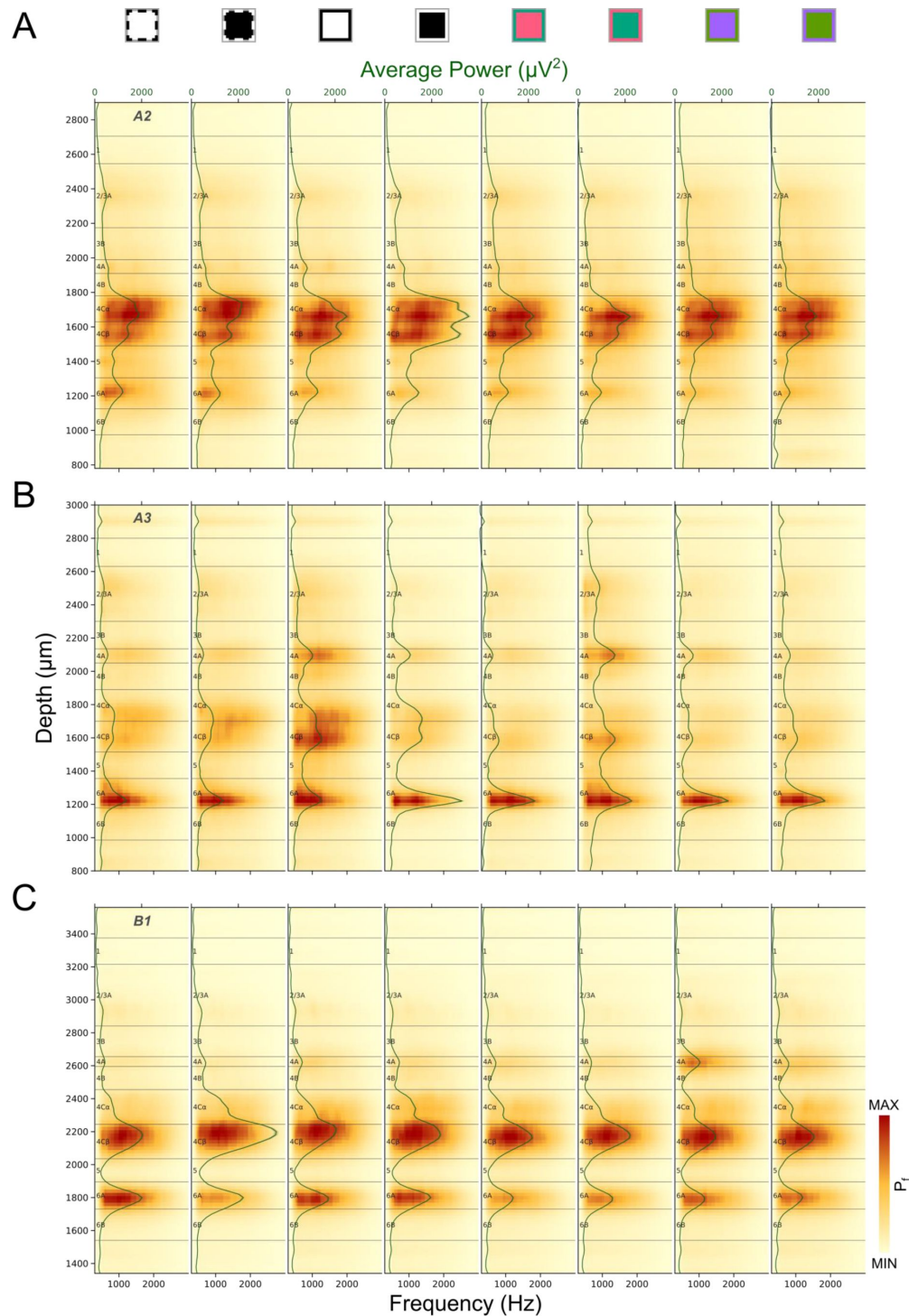
While overall AP power, as described above, clearly generates signals that are reliable and precisely related to cortical layers, differential AP power on different electrode contacts might sometimes result from stochastic variability in the precise locations of electrode contacts in relation to neuronal cell bodies. For example, the AP amplitude measured from a particular neuron is typically much larger on one contact than other nearby contacts (**Figure 4B** [insets](#)). These differences might contribute significantly to differential power signals, suggesting that it might be useful to compare power to an alternative measure that is independent from spike amplitudes. We calculated coherence spectra ( $C_f$ , See Methods) between recording channels to generate a measure that is independent from variation due to different spike amplitudes (**Figure S4A** [insets](#)). Coherence represents normalized phase relationships between two signals, where zero coherence indicates random phase relationships and one indicates deterministic phase relationships. As seen in **Figure 4B** [insets](#), spikes from one neuron usually spread to a group of nearby recording contacts with very small time delays and much attenuated amplitudes. These relationships are expected to result in significant local coherence between neighboring channels. Assuming spikes of a neuron spread to at least two adjacent recording channels, then the coherence between the two channels would be directly proportional to number of spikes, independent of spike amplitude. Thus, coherence is expected to be related to power in that both are related to the number of action potentials that are evoked around an electrode contact, but they differ in that coherence is insensitive to spike amplitudes.

**Figure 6** [insets](#) illustrates laminar patterns of coherence spectra during responses to various visual stimuli for 3 electrode penetrations. Like laminar AP power spectra, coherence is generally greatest in L4C and L6A. While there are generally fewer and less distinct laminar transitions for coherence spectra than for power spectra, there are nevertheless sometimes cases where they can resolve uncertainties. In many penetrations coherence is also a useful and clear indicator of the white matter location due to the transition to a minimal value at the L6/WM border.

## AP Power temporal change profile

Another measure that is expected to be closely related to both neuronal density and responsiveness is the AP power change relative to average power in the baseline time window ( $\Delta P/P$ , See Methods). Because this measure subtracts baseline power it is expected to be less sensitive than AP power spectra to variations in spike amplitude. This measure also has the advantage of revealing timing between layers that might result from different response latencies. It can be seen in **Figure 7** [insets](#) that responses in L4C $\alpha$  are typically faster than L4C $\beta$ , consistent with previous findings that the magnocellular LGN afferents and their recipient neurons in L4C $\alpha$  have shorter response delays than for the parvocellular recipient neurons in L4C $\beta$  ([13](#) [insets](#), [21](#) [insets](#)). Different colored visual stimuli also often generate distinctly different laminar and temporal patterns corresponding to L4A, L4C $\alpha$ , or L4C $\beta$ , that also vary between penetrations in their latency and response strength (**Figure S2** [insets](#), **Figure S3** [insets](#)). For example, for each of the three penetrations shown in **Figure 7** [insets](#), the S-OFF parvocellular afferent-recipient L4A is clearly defined by high  $\Delta P/P$  with longer latency than for L4C $\alpha$ , in response to at least one of the colored visual stimuli. But for one penetration (**Figure 7A** [inset](#)) the L4A band is clearest in response to the S-ON stimulus, while for another (**Figure 7C** [inset](#)) it is only observed in response to the S-OFF stimulus. There is also variability between penetrations and stimuli in the temporal sequence observed for activation of the LGN-recipient layers L4A, L4C $\alpha$ , L4C $\beta$  and L6A. This is particularly striking for the responses to the S-ON stimulus for the penetration shown in **Figure 7B** [inset](#), where there are distinct transient increases that clearly and sharply define the borders of each of these layers and which occur first

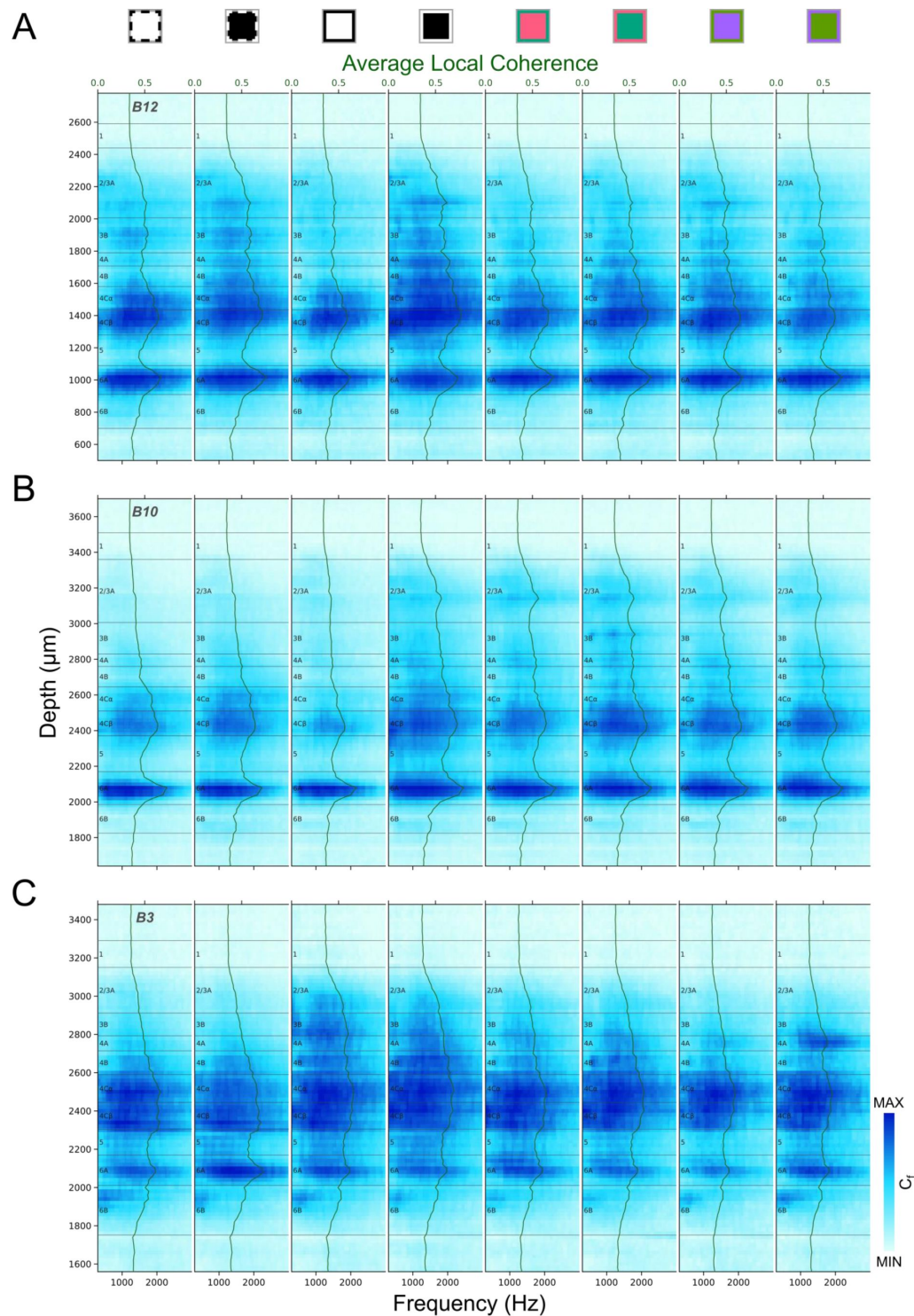




**Figure 5.**

**Laminar power spectrum ( $P_f$ ) profiles for three penetrations (A: A2, B: A3, C: B1) generated in response to different stimulus conditions.**

Power spectrum profiles were independently color-mapped according to the minimum and maximum power in each profile (See [Figure 3](#) legend for stimulus conditions corresponding to icons above plots). The green lines and corresponding shaded ribbons show the Mean $\pm$ SEM power across frequency (300-3000Hz). Note that SEM is roughly the same as the thickness of the line.



**Figure 6.**

**Local coherence spectrum ( $C_f$ ) profiles for three penetrations (A: B12, B: B10, C: B3) generated in response to different stimulus conditions.**

Local coherence profiles were independently color-mapped according to the minimum and maximum local coherence in each profile (See [Figure 3](#) legend for stimulus conditions corresponding to icons above plots). The green lines and corresponding shaded ribbons show the Mean  $\pm$  SEM local coherence across AP frequency (300-3000 Hz). Note that SEM is roughly the same as the thickness of the line.

in L4C $\alpha$ , then L6A, followed by L4A, and finally L4C $\beta$ . More typically, response latencies are similar in L4A and L4C $\beta$ . In all penetrations the responses in L4C $\alpha$  have the shortest latency. Overall,  $\Delta P/P$  profiles often provide laminar information that is more difficult to deduce from other measures, particularly for L4C $\alpha$ , L4C $\beta$ , or L4A.

## Combining Metrics for Accurate, High Resolution Laminar Delineation

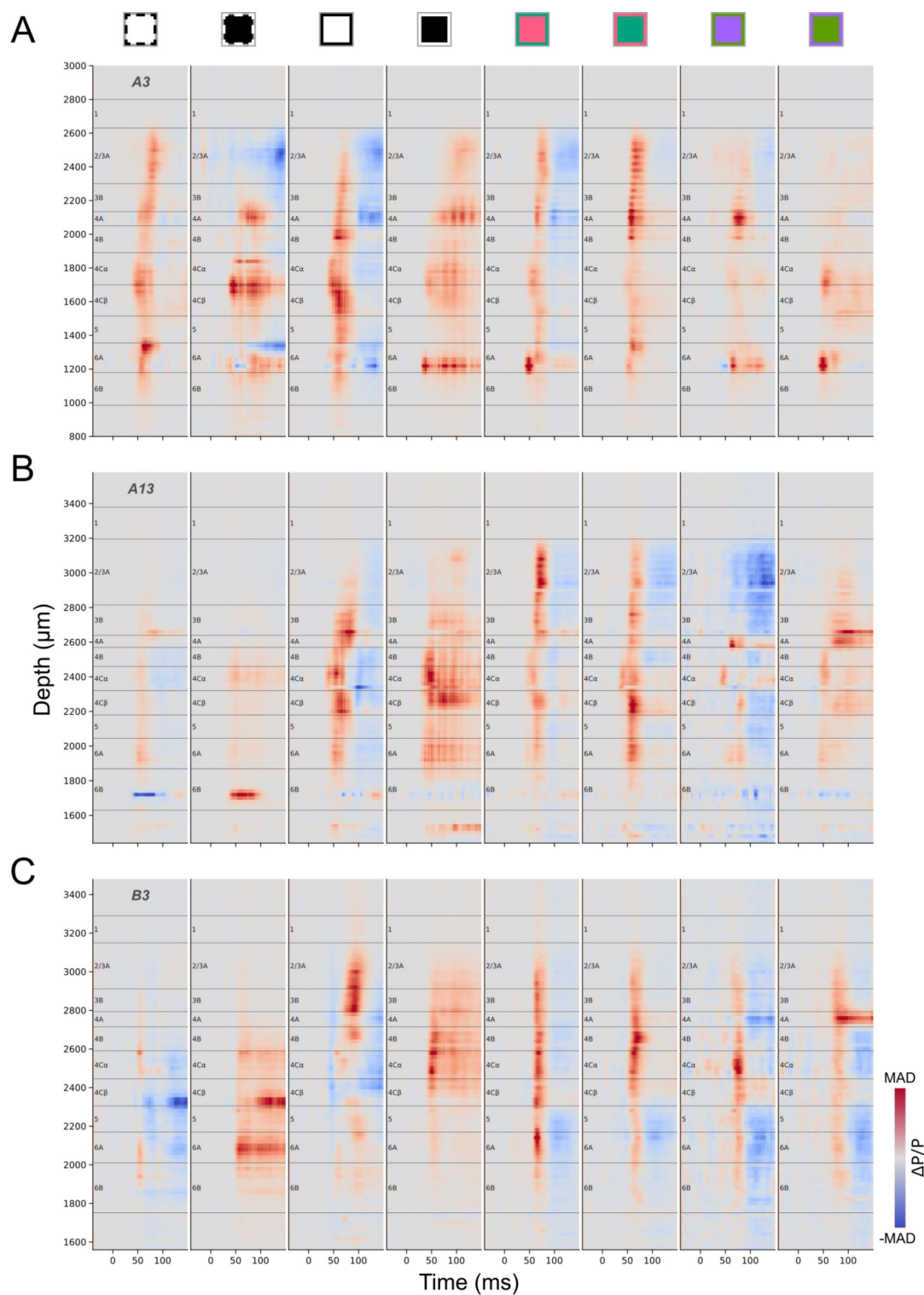
While many of the metrics described above consistently and precisely identify some laminar boundaries, there is not a single metric that can identify all layers. For example, some metrics are most closely related to anatomical properties such as neuron density or soma size, while others reveal differences related to functional properties, such as visual stimulus preference or response latency. In addition, some metrics are very good for providing anchor points, such as the general location of a layer, but are not as good as others for defining precise borders. Finally, the data also demonstrate that there is considerable variability between penetrations, such that the metrics and visual stimuli that best reveal laminar boundaries differ between penetrations. Below, we therefore outline a general order in which various criteria might be ideally applied to accurately identify as many laminar boundaries as possible for a given penetration, but common sense should be used to estimate confidence levels for any particular set of data based on available data quality and consistency.

Below and in [Figure 8](#) we illustrate a general workflow for use of the different metrics, applied to one example penetration. We generally begin by applying metrics that are most consistent and definitive for identification of borders and layers, then constrain possible solutions related to subsequent decisions where individual metrics used in isolation might fail. The particular metrics that are most reliable and which are prioritized differ depending on the particular boundaries that are being evaluated. However, as a general rule, CSD profiles were not necessary for layer identification and mostly were not used, with the exception of occasional cross-referencing of laminar response latencies. Finally, prior knowledge of the range of relative thicknesses and positions of layers based on anatomical studies (e.g. [Figure 1A](#)) is crucial for guiding the steps described below and can often be used to make very close estimates in cases where all metrics combined are still not sufficient to reveal a clear boundary. High resolution layer delineations were all manually derived for each of our penetrations following the guidance below. (See more examples in [Figure S5](#), [S6](#).)

## Top and bottom: L1/L2 and L6/WM

As a first step, it is very helpful to identify the top and bottom of the cortex. Given the known ranges of relative layer depths based on prior anatomical studies, knowledge of the total cortical depth and which electrode contacts are within the cortex provides an important constraint on subsequent possibilities. The L6/WM border can be reliably determined based on several measures that typically all corroborate one another. The most sharp and distinct transitions are the changes in spatial spread, duration, and peak-trough ratio of spikes ([Figure 8F-H](#)). This border can also be confirmed by the decreased coherence at the bottom of L6. Because neuron density increases steeply from L1 to L2, this border also has a clear transition that is most consistent across visual stimuli for relative power change ([Figures 7](#), [8C](#)), but is also clear for multiple stimulus conditions and across metrics based on AP power and coherence ([Figures 5](#), [6](#), [8A](#), [8B](#)). It is not unusual for unit density to fall to zero below the L1/L2 border ([Figure 8E](#)). This is likely because damage caused during initial electrode penetration through the pia can result in poor quality recordings such that units in L2/3A cannot be detected and isolated using spike sorting algorithms. But the other AP band measures (Power and Coherence) clearly show that the corresponding electrode contacts are surrounded by active neurons.

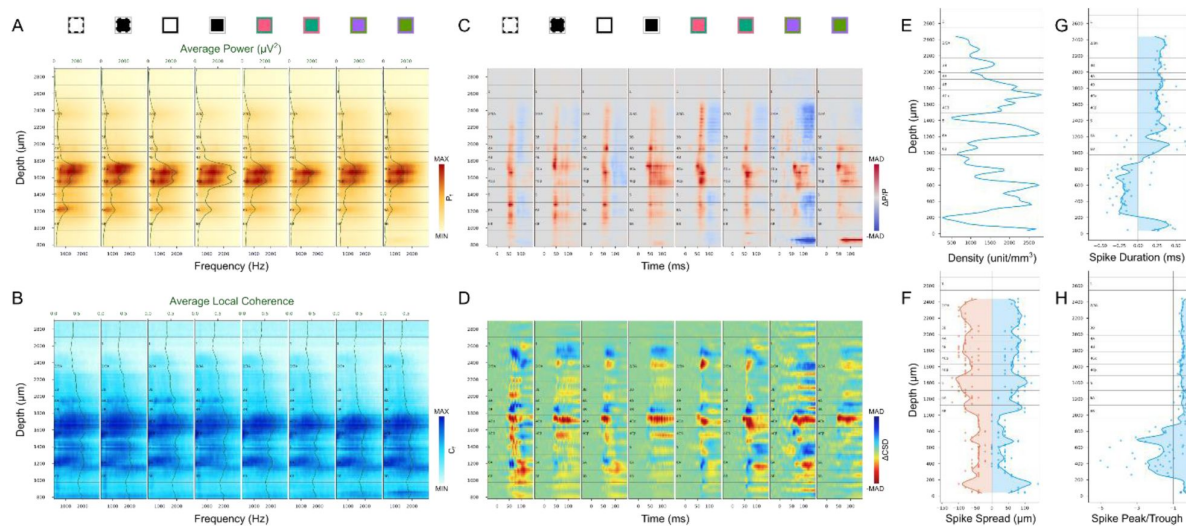




**Figure 7.**

**$\Delta P/P$  profiles for three penetrations (A: A3, B: A13, C: B3) generated in response to different stimulus conditions.**

$\Delta P/P$  (See Methods) profiles were independently color-mapped according to the maximum positive (red) and negative (blue) absolute deviation (MAD) for each profile (See [Figure 3](#) legend for stimulus conditions corresponding to icons above plots).



**Figure 8.**

**V1 layer delineation for an example penetration (A2) based on AP power spectrum (A), local coherence (B), instantaneous AP power changes (C), CSD changes (D), unit density (E), spike spread (F), spike duration (G) and spike peak-trough ratio (H).**

Conventions are the same as for corresponding plots illustrated in earlier figures.



## Deep borders: L4C/L5, L5/L6 and L6A/L6B

The top and bottom of L5, corresponding to the L4C/L5 and L5/L6A borders respectively, are marked by both low anatomical cell density and an absence of LGN input to L5. Functionally, these changes are most consistently and easily recognized as sharp edges in AP power spectra, also corroborated by AP local coherence spectra (**Figure 8A-B**). L5 neurons also have the largest soma size in cortex that can be corroborated by the spike spread distribution (**Figure 8F**). The L6A/L6B border is identified from the decreases in AP power and local coherence that are found about halfway from the L5/L6A border to the L6/WM border. This relative width matches closely to the change in neuronal density that is used to define the L6A/L6B border in Nissl-stained anatomical specimens and also matches closely with the termination of magnocellular LGN afferents in L6B (12).

## Layer 4 sublayers: L4A, L4B, L4C $\alpha$ , and L4C $\beta$

L4B, like L5, is a relatively cell sparse zone that lacks direct LGN input. Accordingly, the same criteria that distinguish the top and bottom borders of L5 can usually be used to identify the L4B borders. In particular, the L4B/4C border is almost always distinguished by a sharp decrease in AP power and local coherence (**Figure 8A-B**). Since L4B neurons are the projection targets of L4C neurons, their latencies are expected to be longer than that of L4C and sometimes this transition can be identified in  $\Delta P/P$  profiles (**Figure 7A**, **8C**). In the great majority of penetrations there is a clear thin band of signal corresponding to the expected location of L4A for at least one of the AP metrics under one of the stimulus conditions (e.g. **Figure 8C**). The top of this band is marked as the L3B/L4A border and the bottom as the L4A/L4B border. In 4 of 31 penetrations, no such electrical signature was seen to indicate the position of L4A. In these cases, L4B can be estimated to be half the thickness of L4C, and L4A can be estimated to be 50 $\mu$ m thick.

Anatomical studies reveal a clear change in neuronal density at the L4C $\alpha$ /L4C $\beta$  border and the thickness of L4C $\alpha$  is invariably at least as large as for L4C $\beta$ . The electrical signatures that best identify the L4C $\alpha$ /L4C $\beta$  border are due to the latency differences expected from the faster conduction velocity for magnocellular afferents to L4C $\alpha$  versus parvocellular to L4C $\beta$ . Between the metrics that reveal latencies,  $\Delta P/P$  is more reliable and yields sharper transitions than  $\Delta CSD$  profiles. When there is a clear latency transition in the middle of L4C (e.g. **Figure 7A**, **7B**, **8C**, **8D**) this is usually marked as the L4C $\alpha$ /L4C $\beta$  border, but the position is always placed such that L4C $\beta$  is not thicker than L4C $\alpha$  because this is a consistent feature in anatomical observations. When there is no observable latency difference, the L4C $\alpha$ /L4C $\beta$  border is placed halfway between the top and bottom of L4C.

## Superficial borders: Pia/L1 and L2/3A/L3B

There are no clear electrical signals to distinguish the precise borders between the pia and L1 or between L2/3A versus L3B. Because L1 contains only a very low density of inhibitory neurons, it is often the case that few or no units are isolated above the L1/L2 border. Nevertheless, it is clear that no units can exist above the pial surface, so any neuron that is isolated above the L1/L2 border can be confidently assigned to L1. For simplicity, we simply assign the L1/pia border to a position ~120-150 $\mu$ m above the L1/L2 border, as this is typical in postmortem histology and invariably incorporates any neurons that are recorded in L1.

The large span from the bottom of L1 to the top of 4A (L2/3) is relatively uniform in histological preparations and is often considered as a single layer. Nevertheless, there are clear differences in the connections and functional properties of neurons in the upper third of L2/3 versus the bottom third. For example, the CO staining intensity and the density of koniocellular S-ON afferent axons are both greater at the bottom of L2/3 (12, 14) and the local axonal projections of L4C neurons rarely extend above the bottom third of L2/3 (22, 23). While none of these

differences relates to a distinct border, it is useful to identify neurons that are more superficial versus deep in L2/3, and we therefore simply mark the border between L2/3A and 3B at a position two-thirds of the distance between L1/L2 and the top of 4A.

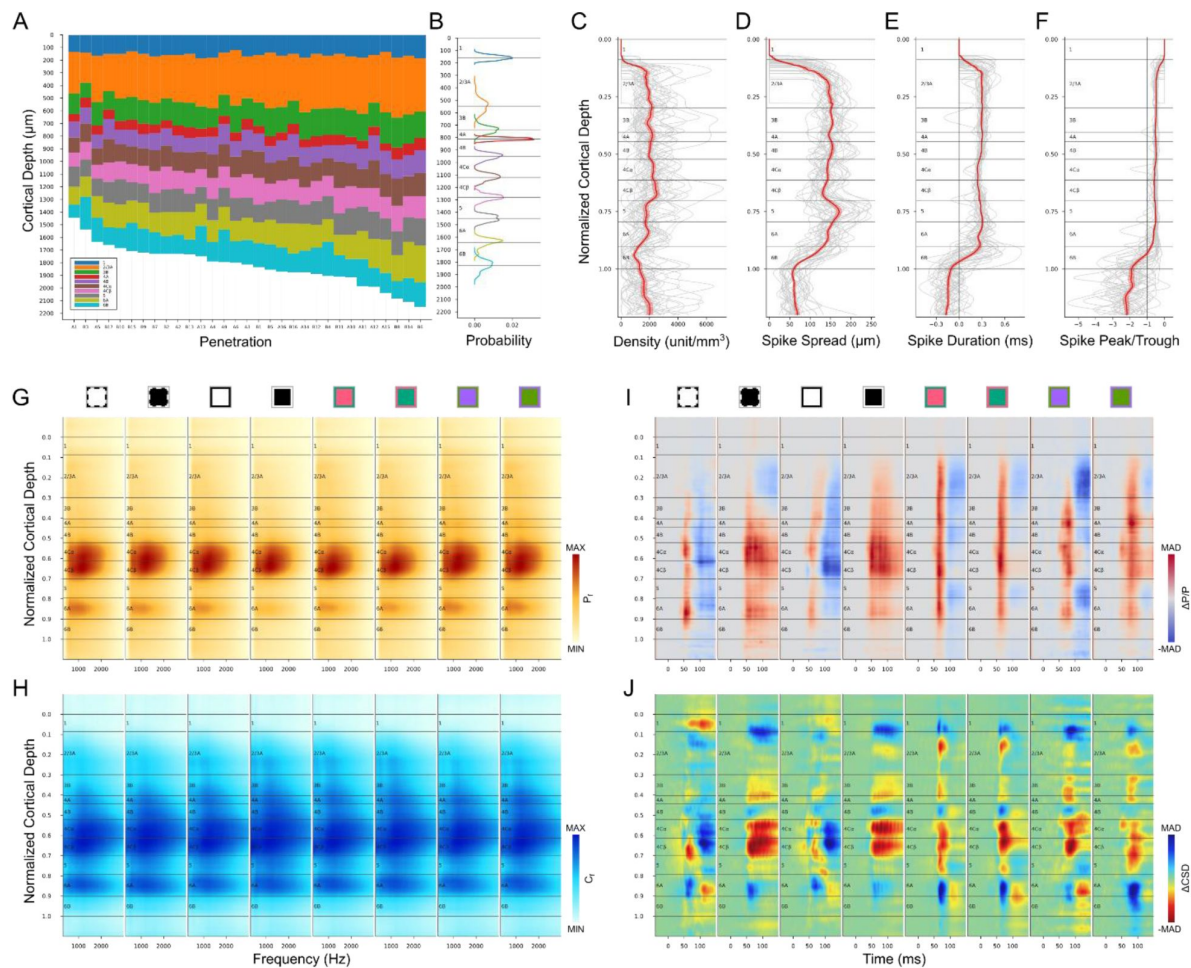
## Average metrics of V1 layer template

**Figure 9** [↗](#) illustrates the laminar boundaries and thicknesses derived for all 31 electrode penetrations in our sample. Note that the overall thickness of the cortex in our physiological penetrations is somewhat more variable than is seen in postmortem histology, ranging from about 1.5 to 2.0mm. This is expected because there can be expansion due to brain swelling or shrinkage due to the placement of a coverslip onto the surface of the cortex in order to generate pressure and minimize brain movement; these influences can vary over the 4-5 days during which recordings were made from each animal. These influences are generally expected to generate proportional changes in laminar thickness across the cortical depth. An additional factor is that cortical thickness is known to decrease closer to the V1/V2 border and this is also seen in our data set (**Figure S7A** [↗](#)). Otherwise, there were no significant differences between monkeys or between ISI-identified functional domains (**Figure S7B-D** [↗](#), Kruskal-Wallis Test,  $p > 0.05$ ). We used the complete set of 31 penetrations to make a proportional alignment and generate an “average” V1 layer template based on the average thickness of each layer (**Figure 9B** [↗](#)). This can be useful when laminar boundaries must be estimated in the absence of empirical data.

We also generated averaged depth distributions for each metric (See Methods). The overall shape of the average unit density distribution (**Figure 9C** [↗](#)) matches that of the anatomical neuron density distribution reported in **figure 6** [↗](#) of the study from Kelly and Hawken (18 [↗](#)), with highest density in L4C $\beta$  and decreased density in L5. Unit density also increases at the L5/L6 border, and in L4A compared to L4B and L3B. Due to the limited sampling volume of a linear probe, tissue damage, and spike sorting capabilities, the unit densities are far less than neuron densities calculated from immunohistology, and density contrast across layers is greatly reduced. The average spike spread (**Figure 9D** [↗](#)) matches the histological observation that L5 neurons have the largest cell bodies while L4C $\beta$  neurons are the smallest. Note that the decreased spike spread in L4A is also expected from the small cell bodies in that layer. The location of the white matter boundary has a marked drop in spike spread corresponding to the transition from spikes recorded from cell bodies versus axons. This transition can also be seen in average distributions of spike duration (**Figure 9E** [↗](#)) and spike peak-trough ratio (**Figure 9F** [↗](#)).

The average power spectrum (**Figure 9G** [↗](#)) and local coherence (**Figure 9H** [↗](#)) distributions clearly show the expected decrease in L5 relative to L4C and L6A. The pattern was also stable across different stimulus conditions implying closer relation to anatomical properties than functional ones. The average relative power change (**Figure 9I** [↗](#)) shows strongest activation in L4A using S/(L+M) cone-opponent colors. It also can be seen that the latencies of 4Ca responses were shorter than for 4C $\beta$ . Although CSD was rarely used in our layer identification process, the average  $\Delta$ CSD profiles (**Figure 9J** [↗](#)) were not far from the expectations based on many previous results. It seems that switching from white to a black screen evoked the strongest increase of current sinks in L4C. For all stimuli, L5 shows increased current sinks while L6A shows increased current sources. L4B shows increased current sources, while L4A/L3B shows increased current sinks.

## Discussion



**Figure 9.**

### V1 layer template and average metrics across cortex.

(A) Identified layers for 31 electrode penetrations aligned to the surface of cortex. (B) V1 layer template and probability density function for the thickness of each layer, aligned on the corresponding boundaries from the layer template. (C-J) Average depth distributions from 31 electrode penetrations. (C-F) First group of identification metrics (C: unit density, D: spike spread, E: spike duration, F: spike peak-trough ratio) on the normalized layer template (gray lines for each penetration, red line and shaded ribbon for Mean $\pm$ SEM). (G-J) Second group of identification metrics (G: power spectrum, H: local coherence spectrum, I:  $\Delta P/P$ , J:  $\Delta CSD$ ) on the normalized layer template. Conventions are the same as in previous figures.

## Laminar identification in primate V1

We have described a set of metrics based on high-resolution AP-band (0.3 – 10kHz) signals from Neuropixels probes that can be used to identify cortical layers in primate V1 at higher resolution and with greater accuracy than using LFP band signals (0.5 – 500Hz) and CSD analysis. Our methods capitalize on properties of electrical signals that can extract three key features of V1 laminar organization: 1) anatomical differences in the size and density of cell bodies; 2) functional differences in the LGN afferents and recipient neurons in different layers; 3) laminar termination patterns of functionally distinct LGN afferents. Accordingly, anatomical differences can often be extracted even without the use of visual stimuli, while functional differences require activation with different visual stimulus conditions and often also incorporate consideration of response latencies.

## Limitations of LFP signal and CSD analysis

Here, we have shown that very close (20 $\mu$ m) electrode spacing and increased electrode density does not necessarily result in higher resolution identification of laminar boundaries based on CSD profiles. Nevertheless, low density electrode arrays and CSD methods have been used for cortical laminar identification in various species and cortical areas (24, 25), in both awake (8) and anesthetized recordings (9). Although the use of CSDs has evolved from simple second spatial derivative to model based inverse methods (7, 26, 27), the underlying assumptions of the forward problem remain the same: continuous CSD distribution, quasistatic electrodynamics and ohmic extracellular media (6). When electrode spacing is close to or less than 20 $\mu$ m, the assumption of continuous and macroscopic CSD distribution might not be valid, and models of CSD distribution based on discrete microscopic sinks and sources may be more appropriate. The alternating sink and source patterns of CSD profiles (**Figure 2A**) derived from 20 $\mu$ m spacing LFPs may reflect the microscopic CSD dipoles of neurons. Thus, an appropriate spatial averaging or down sampling of electrodes was needed to produce smooth, useful CSD profiles in which regions of sinks and sources can be potentially related to cortical layers or sublayers. The sizes of smoothed sink or source regions are typically most suitable for matching the thickness of layers and sublayers when electrode spacing is around 50 $\mu$ m. This implies that further reductions in probe spacing will not improve the ability of CSDs to identify cortical layers.

The CSD profile represents spatial and temporal patterns of electrical activities across layers, thus it naturally depends on the structure and dynamics of functional cortical circuits. We have shown that for any given electrode penetration, differential activation of geniculocortical pathways using different colors or achromatic stimuli generates distinct CSD profiles. For example, CSD profiles generated in response to S-cone stimuli can have relatively large sinks (depolarization) in layers 4A and 3B, similar to those generated by directly activating koniocellular LGN neurons using electrical and optogenetic stimulation (9). CSD profiles generated from black and white stimuli also show dramatic differences. Flipping the screen from white to black consistently generated strong early sinks that roughly matched layer 4C. On the contrary, flipping from black to white sometimes generated early sources (hyperpolarization) instead of sinks (**Figure S2D**) making it less reliable than a black stimulus. The large white stimulus has been reported to induce sustained inhibition which was absent in black stimulation (20). This suggests that CSDs from black and white stimuli should be analyzed separately for layer identification.

To increase spatial resolution of LFP signals, prior studies have restricted LFP analysis to the gamma band (30-100Hz), which should theoretically spread over shorter distances than when lower frequencies are also included (17). Indeed, when this is used to identify monkey V1 layers, a layer 4C sink is detected with a somewhat sharper cutoff at the likely L4C/L5 border than for the CSD (10). When we apply this analysis to our data collected with Neuropixels probes (**Figure S8A, C**), we see similar patterns to those previously reported (10); however, the



ability to accurately detect laminar boundaries or to definitively identify the precise borders of layer 4C are not comparable to what can be achieved using AP band signals from high-density Neuropixels probes.

LFP gamma band coherence has also been used ([10](#), [24](#), [28](#)) to categorize ensembles of neuron populations, and several of them have been shown to be confined to specific layers. We analyzed local coherence using gamma LFP (**Figure S8B, D**) and find that the ability to detect laminar boundaries is also much poorer than signals based on AP band coherence, even when using high density Neuropixels probes. It is unclear what mechanisms generate the laminar coherence in gamma LFPs, however, our clearer laminar signals based on AP band local coherence indicates that this is likely due to the sharper falloff of high frequency signals, which can be exploited using only the densely spaced Neuropixels electrode contacts.

## AP signals have high spatial resolution

In contrast to the inherent low resolution of LFP-based signals, signals based on AP band metrics have the potential to allow higher resolution laminar detection when using high-density probes. The various AP-band metrics that we have extracted and used to allow laminar identification include: unit density, spike spread, duration and peak to trough ratio, power spectrum, local coherence spectrum and temporal power change profile. Most of these measures could be analyzed based on spontaneous activity and therefore might be utilized in brain areas other than V1, such as higher cortical areas where the additional signals derived based on visual response latencies or laminar differences in functional input properties are not as prevalent. Indeed, in rat hippocampus, it has been demonstrated that high frequency power ( $\geq 300\text{Hz}$ ) can be used to differentiate different layer structures ([28](#)).

We have described the hypothetical generative models of AP band power and local coherence based on superposition of individual spikes. In the case of similar order of firing rate across layers, the depth distributions of AP band power and local coherence mainly represent depth distributions of neuron density which is the most definitive anatomical property for layer delineation. Therefore, AP band power and local coherence should be much less dependent on stimulus, as demonstrated in our results. Moreover, they remained largely unchanged even in the baseline period (**Figure S9**) in which cortical circuits were believed to have been stabilized following stimulus-induced activity. Since AP power and local coherence were homogenous across frequencies, they could also be first averaged across frequency, then shown for consecutive trials (**Figure S10**). This opens the possibility of online layer estimation without visual stimulation.

The improvements in layer identification that we have demonstrated using high-resolution AP-band signals and high-density probes suggest that still greater improvements might be possible. The estimation of neuron density profile, for example, should be more accurate when more neurons can be detected and isolated by spike sorting. This can be achieved by decreasing electrode spacing or by using multiple shanks. Neuropixels 2.0 probes ([29](#)) have even closer electrode spacing at  $15\mu\text{m}$  and up to four shanks; thus, the spike-sorted units within and across layers should allow even more accurate estimation of neuron density distribution and better laminar delineation. Furthermore, new development of spike sorting and drift correction algorithms ([30](#)) that incorporate detailed modeling of spike spread could better estimate the three dimensional positions of units, leading to better unit identification and drift estimation, and may also estimate soma properties, giving additional anatomical information useful for laminar differentiation.

In addition to stimulus independent measures, we have also incorporated responses to visual stimuli for our laminar analyses based on power spectra, local coherence spectrum and stimulus-evoked temporal changes in AP power. We have demonstrated the ability to detect functional distinctions that are peculiar to monkey V1. For example, we found that the most effective way of revealing the thin layer 4A was to use colors that modulate only LGN S Cone inputs. The response

latency differences between 4C $\alpha$  and 4C $\beta$  are also useful to separate these two otherwise ambiguous layers. On the other hand, the L-M cone opponent colors were expected to much more strongly activate layer 4C $\beta$ , however, substantial activation with little delay was observed across supragranular, granular and infragranular layers, implying complex intralaminar processing of color in V1 (19 [↗](#)).

## Consideration of electrode drift

There are several factors that can cause the relationships between electrode contacts and laminar borders to change over time. These include both changes on a fast time scale (such as head movement, heart rate, or breathing) and slower drift, such as expected from brain swelling. In addition, there can be either “rigid” uniform movements that simply shift the locations of contacts relative to the tissue, or non-uniform changes that might differentially affect the relationships at different locations on the probe (such as compression of superficial layers during initial introduction of an electrode).

In our experiments, electrode movement was minimized and often not detected, resulting in stable recordings (Figure S12A [↗](#)). This is primarily because we used light compression of the tissue by placement of a coverslip at the surface, which largely eliminates fast movements from heartbeat and breathing. Furthermore, the animals are anesthetized and paralyzed, preventing changes due to head movements. Nevertheless, slow drift was detected in some of our recordings, likely due to swelling of brain tissue over time (Figure S12B [↗](#)). We corrected for drift using Kilosort (31 [↗](#)) version 3 on concatenated AP data produced by CatGT (See Methods). This approach is based on signals in the AP band which, as described above, are expected to have better spatial resolution than LFP signals.

In contrast, prior studies have used LFPs to track probe motion and in turn correct for motion in human cortex (32 [↗](#)). However, LFPs may not be a stable or accurate indicator of electrode drift because the depth pattern of LFP might be changed by external stimuli or internal spontaneous activities, dissociating the temporal change of LFP patterns from probe motion. On the other hand, spike sorting algorithms (e.g. Kilosort 3.0) should be able to use spike amplitudes to track tissue displacement relative to probe contacts with higher accuracy than LFP. While this requires sampling sufficient numbers of spikes, introducing tradeoffs between spatial resolution and sensitivity, these tradeoffs are minimized by the use of high-density electrode arrays, such as Neuropixels.

Regardless of whether laminar boundaries are estimated from CSDs, LFPs, or AP signals, matching unit positions to laminar locations should use motion correction for both rigid and non-rigid sources. We found that strict reliance on a rigid model for motion correction with Kilosort version 3.0 failed to accurately identify slow drifts in some of our penetrations (Figure S12E [↗](#)). We therefore also introduced additional methods to allow for non-rigid drift (See Methods).

## Towards a universal in vivo pipeline for cortical layer identification

In this study, we developed AP metrics that can be used to estimate several important anatomical and functional properties of cortical layers. To accurately identify each layer, multiple metrics need to be combined, since no single metric is sufficient to reveal all layers. Here, layer boundaries were manually drawn, but automatic/semi-automatic algorithms can be constructed. Template matching algorithms have been developed to automatically align probe electrodes to cortical layers (33 [↗](#)). Since multiple metrics would need to be incorporated, complicated regression algorithms would potentially be needed to satisfy multiple objectives, and the layer thickness template we derived (Figure 9B [↗](#)) could be used as a prior to penalize large deviations from the template. On the other hand, supervised and unsupervised machine learning algorithms

might be able to fully exploit all metrics and provide less biased estimates (34). However, large datasets and annotations may be needed to train and test the artificial neural network models to ensure that they can operate at an accuracy level no less than human operators.

The laminar organization of cortical circuits is thought to be related to functional circuit mechanisms underlying many cognitive functions. Thus, assignment of in vivo recorded neurons to cortical layers could facilitate anatomical and functional analysis of information processing in cortical circuits in areas beyond the monkey primary visual cortex. The expectation that evoked initial sinks correspond to input layer 4 is likely limited to primary sensory areas where the main feedforward input from the thalamus terminates densely in that layer and can be driven robustly by a single stimulus. It has been shown that the location and thickness of initial sinks could vary across multiple mouse visual cortices, making it hard to identify layer 4 according to initial sinks (35). Moreover, in cortical areas that do not directly process sensory information, such as prefrontal cortex, the input layer 4 may not be easily activated by external stimuli, rendering CSD methods unusable. The AP metrics estimating mainly anatomical properties could overcome the structural and functional variations across different cortical areas and species, leading to a general solution for in vivo layer delineation. Future simulation studies based on biophysically plausible cortex models would not only reveal the applicable conditions and limitations of the AP metrics we described, but could also lead to identification of new metrics to more accurately estimate essential anatomical properties, such as neuron density and soma size/shape, that have been accepted for universal delineation of cortical layers.

## Materials and methods

### Animal Surgery

Two rhesus monkeys (*M. mulatta*, adult, male) were used for the acute experiments. All procedures using live animals were approved by the Institutional Animal Care and Use Committee (IACUC) at The Salk Institute for Biological Studies. Detailed surgical procedures have been described previously (36).

Animals were injected intramuscularly with atropine sulfate (0.02 mg/kg), and then anesthetized with ketamine (10 mg/kg). During surgery, animals were kept under anesthesia (1-2.5% isoflurane in O<sub>2</sub>) and sterile conditions. Tracheostomy was performed, and animals were mechanically ventilated. EKG, SpO<sub>2</sub>, end tidal CO<sub>2</sub>, and body temperature were all monitored throughout the experiments. Lactated Ringer's with 5% dextrose was given intravenously. Antibiotics (Cefazolin, 25 mg/kg, i.v.) were administered every 12 hours. Mannitol (20%, i.v., 200mg/kg) was given slowly 30 minutes before craniotomy was performed. Dexamethasone (1-2 mg/kg, i.v.) was given daily.

A large craniotomy (~25mm diam.) and durotomy were performed over the primary visual cortex, and a customized recording chamber was cemented onto the skull. A coverslip was mounted to reduce brain movement. Two silver wires were implanted from which EEG signals were monitored to assess anesthetic depth. Anesthesia was gradually transitioned from isoflurane to sufentanil citrate (2.0-8.0 µg/kg/hr, i.v.), and vecuronium bromide (0.05-0.2 mg/kg/hr i.v.) was administered to paralyze the animal after the chamber was implanted. Sufentanil dosage was adjusted to ensure slow wave EEG.

### Extracellular Electrophysiology

A Neuropixels 1.0 probe (IMEC, Leuven) was held by an oil hydraulic manipulator (Narishige International USA, Inc.), which was attached to the chamber. The probe was aligned perpendicular to cortex and slowly inserted around 3mm below brain surface through a small hole (~1mm diameter) in the coverslip. Then, the hole was covered with saline to prevent drying. Reference and Ground of the probe were soldered together and connected to the chamber. Extracellular

electrical potentials from 384 channels were sampled at 30kHz and streamed to a computer through the Neuropixels acquisition module (IMEC, Leuven) within a PXIe system (PXIe-1071 chassis, PCIe-8381, PXIe-8381, PXIe-6341, National Instruments, Corp.). Data acquisition was controlled by SpikeGLX (<https://billkarsh.github.io/SpikeGLX>). Raw data were filtered and saved separately to an action potential band (AP band, 0.3 – 10kHz, 30kHz) and a local field potential band (LF band, 0.5 – 500Hz, 2.5kHz). Recordings started around 30 minutes after probe insertion when responsive visual fields were manually mapped. The time interval between different penetrations of the same animal was at least 3 hours.

## Visual Stimulus Presentation

Visual stimuli were generated and presented using Experica (<https://experica.org>) on an LCD monitor (ROG SWIFT PG279Q, Asustek Computer Inc.) with a refresh rate of 144Hz. The monitor was positioned 57cm in front of the animal's eyes, which were dilated with 1% atropine sulfate and 0.5% proparacaine hydrochloride ophthalmic solution. Both eyes were fitted with contact lenses to focus on the screen.

Prior to each experiment, the luminance of the red, green and blue primary colors of the monitor were measured using a spectroradiometer (PR-701, Photo Research, Syracuse, NY) and linearized by a color lookup table. Then, the spectra of red, green, blue primary colors were measured at resolution of 2nm. Three color pairs on the three axes of DKL color space with maximum achievable cone contrast: L+M luminance (white/black), L/M cone opponency (red/green), and S/(L+M) cone opponency (purple/lime), were calculated (<https://github.com/Experica/ColorLab.jl>) based on the spectra of red, green, blue primary colors and the 10° cone fundamentals (37). The two colors of a pair were presented alternatively, each for 1.5 second on full screen and repeated 100 times. All the color pairs were presented to dominant eye, whereas only the L+M luminance color pair was presented to non-dominant eye. The total time of presenting all stimuli described here was in 30 minutes or less.

## Data Analysis

Data were first processed (<https://github.com/Experica/NeuroAnalysis>) in MATLAB (Mathworks Inc., MA), then analyzed (<https://github.com/Experica/NeuroAnalysis.jl>) in Julia (<https://julialang.org>).

In order to correct probe drift during whole recording session for each electrode penetration and assign consistent ID for each unit in spike sorting, all AP band data of one penetration were concatenated in time order by CatGT (<https://billkarsh.github.io/SpikeGLX>). In the process, a subsampling method was used to demultiplex (TShift option) channels that shared the same ADC of the probe, then a 300Hz high pass Butterworth filter of order 3 was applied to remove offset, and finally a global common average referencing was used to remove shared noise.

## Drift correction and spike sorting

Kilosort (31) version 3 (<https://github.com/MouseLand/Kilosort>) was used on concatenated AP data produced by CatGT, and the sorting results were curated in Phy (<https://github.com/cortex-lab/phy>). Prior to spike sorting, whitened data were saved to a separate file and a rigid registration method was used to correct probe drifting in the file. In the anesthetized state, most of our penetrations showed little to no drift.

For a few penetrations in which significant drift occurred, non-rigid drifting was found prominent, and a non-rigid registration method (imregdemons in MATLAB) was used instead of the rigid one in Kilosort if the rigid drift correction result was not satisfactory (Figure S12). The putative double counted spikes (inter-spike-interval  $\leq 0.35$ ms) within a cluster and across nearby



(within circle of 60µm radius) clusters were removed before manual curation in Phy. Clusters with low mean firing rate ( $\leq 0.5$  spike/second) were regarded as noise. On average, around 200 single-unit or multi-units can be resolved for one penetration.

## Spike waveform feature and unit density

The spike waveforms of a unit were calculated by averaging randomly chosen spike AP segments (<1000) of the unit. The heights (peak subtracts trough) of spike waveform on different channels were used as weights to estimate (center of mass) unit position. The channels used in weighted average of channel positions were chosen within a circle of 65µm radius, centered on the channel of the largest spike height. The upward and downward spread of spikes were defined as the largest vertical distances between the channel with the maximum spike height and the closest channels with spike height less than 15% of the maximum spike height (38 [Fig. 3](#)). Using the spike waveform of maximum height, the spike duration was defined as the time of peak subtracting time of trough, and spike peak-trough ratio was defined as the ratio of peak and trough amplitudes.

The unit density was estimated by assuming a 100µm radius cylinder as the unit detection volume of the probe. Depth distribution of unit density was derived by counting number of units in a 40µm segment of the cylinder moving along the probe with 20µm step and then smoothed by a gaussian filter ( $\sigma=28\mu\text{m}$ ). A single unit was counted as one neuron, whereas a multi-unit was counted as 1.2 neurons. This value was empirically chosen based on a general impression of contamination rate of multi-units and likelihood of sharing the same units among adjacent multi-units. Since the percentage of multi-units was low (~20%) in most of our penetrations, varying this value from 1 to 2 did not substantially change the signature shape of unit density distribution. The same methods of moving window and smoothing in unit density estimation were also used to estimate depth distributions of upward spread, downward spread, spike duration and spike peak-trough ratio.

## Current source density

The analysis time window of each stimulus presentation was chosen as [-40, 150] millisecond around stimulus onset time 0, where [-40, 10] was used as baseline window and [10, 150] as response window. These definitions also apply for the power spectrum, local coherence spectrum and temporal power change described subsequently. The local field potentials in analysis window were first notch filtered to remove electrical line noise (60Hz), then bandpass filtered in 1-100Hz and down sampled to 1kHz. LFPs with the same depth were first averaged while excluding the reference channel, then trial averaged to produce a series of LFP traces across cortex with a depth resolution of 20µm. Current source densities were calculated for each stimulus from the series of LFP traces. Specifically, the inverse CSD method (7 [Fig. 3](#)) was used, and the

$$\Delta\text{CSD}(t) = \text{CSD}(t) - \overline{\text{CSD}_b} \quad (1)$$

current source densities were modeled as infinitely thin disc with 500µm radius. The stimulus induced CSD changes ( $\Delta\text{CSD}$ ) were defined as CSD subtracting the temporal average CSD in the baseline time window.

## AP band power and coherence spectrum

In order to keep aligned with positions of sorted units, analysis was based on the file containing whitened, drift-corrected AP band data produced by Kilosort before spike sorting. The AP band data were unwhitened, bandpass filtered in 300-3000Hz, and down sampled in 10kHz. Power spectral densities across frequencies ( $P_f$ ) in the response window ([10, 150]ms) were estimated by multi-taper method. The instantaneous

$$\frac{\Delta P}{P}(t) = \frac{P(t) - \overline{P_b}}{\overline{P_b}} \quad (2)$$

powers in analysis window ([−40, 150]ms) were evaluated by mean of squares ( $\text{RMS}^2$ ) in a 10ms sliding window. In the same way as for LFP, power spectral densities and instantaneous powers were averaged across same depth and trials. Depth distribution of power spectrum was smoothed by gaussian filtering ( $\sigma=20\mu\text{m}$ ). The stimulus induced instantaneous power changes ( $\Delta P/P$ ) were defined as relative change to average instantaneous power in the baseline window ([−40, 10]ms).

Coherence is defined as follows:

where  $S_{ii}(f)$  and  $S_{jj}(f)$  were power spectral densities of channel  $i$  and  $j$  respectively, and  $S_{ij}(f)$  was cross power spectral density of channel  $i$  and  $j$ . The local coherence spectrum ( $C_f$ ) for each channel was evaluated

$$C_{ij}(f) = \frac{|S_{ij}(f)|^2}{S_{ii}(f)S_{jj}(f)} \quad (3)$$

using weighted averaging where pairing channels were confined within a circle of  $55\mu\text{m}$  radius, and distance between pairing channels was used to obtain gaussian weight ( $\sigma=25\mu\text{m}$ ). Coherences in the response window ([10, 150]ms) were calculated by multi-taper method and then averaged across same depth and trials as described for LFPs to produce the distribution of local coherence spectrum along the probe.

## Layer normalization template

Several penetrations were excluded from the dataset when: (1) [the health of the cortex had deteriorated](#), usually due to prolonged duration of maintenance under anesthesia (A17, B18); (2) [the penetration was very close to the V1/V2 border](#), where the whole thickness of cortex is thinner than other regions of V1 (A7); (3) [an error was made in assignment of the dominant eye](#), resulting in lack of data under several stimulus conditions (A8). The average thickness of each layer was used to build the V1 layer template, then the thickness of each layer for each penetration was linearly scaled to match the thickness of the corresponding layer in the V1 layer template. After mapping of depth coordinates of each penetration to the layer template, the depth distributions of identification metrics were interpolated (Cubic Spline) in depth coordinates of the template so that the average metrics on aligned layer template can be calculated. For power spectrum, local coherence spectrum, instantaneous power changes, and  $\Delta\text{CSD}$ , profiles of each stimulus presentation for each penetration were normalized before averaging across penetrations.

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### Author contributions

Conceptualization: LZ, PL, EC

Data Collection: LZ, PL

Methodology: LZ

Writing: LZ, PL, EC

### Competing interests

Authors declare that they have no competing interests.

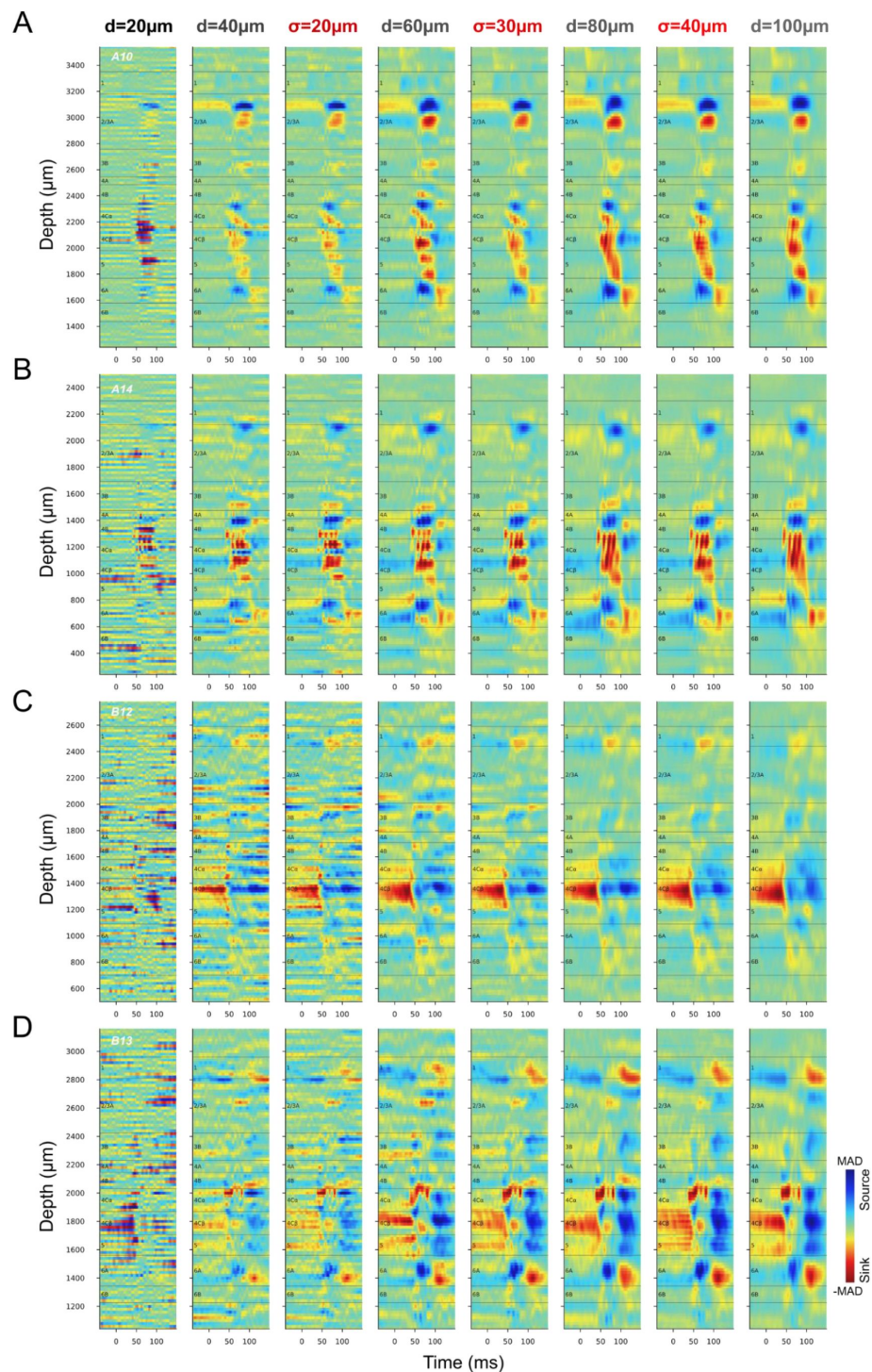
### Data availability

All data necessary to support the paper's conclusions are present in the main text, supplemental information, or available from the corresponding author upon reasonable request. The results presented here are based on more than 5TB of raw data and include data measuring activity during stimulus conditions that are not relevant to the present manuscript. The authors are presently conducting further analyses of that data which will be published at a later time. These data are not structured to allow data relevant to this manuscript to be separated from that which is reserved for future publication.

### Code availability

Customized code used to perform the analyses can be found at [https://github.com/babaq/EPhys\\_V1\\_Layer](https://github.com/babaq/EPhys_V1_Layer)  and <https://github.com/Experica> .

## Supplementary figures

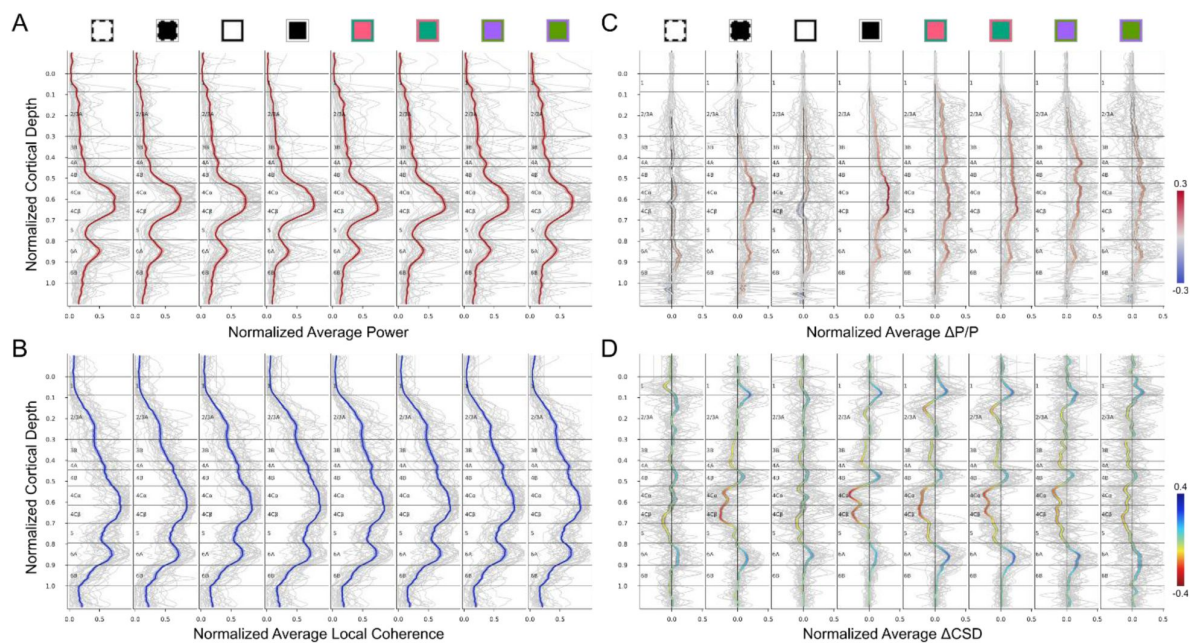


**Figure S1.**

**Comparing spatial averaging to downsampling for CSD profiles evoked by flipping the black screen to the white screen through the dominant eye.**

The same four penetrations in [Figure 2E-H](#) are shown here (A: A10, B: A14, C: B12, D: B13). In the downsampling process, LFPs from electrodes with increasing vertical spacing ( $d=20, 40, 60, 80, 100\mu\text{m}$ , gray) were used to calculate CSDs, then they were interpolated (Cubic Spline) to the vertical resolution of 20 $\mu\text{m}$ . In the spatial averaging process, CSD profiles of 20 $\mu\text{m}$  spacing were Gaussian filtered with increasing width ( $\sigma=20, 30, 40\mu\text{m}$ , red). The CSD profiles were independently color-mapped according to the maximum absolute deviation (MAD) in each profile.





**Figure S2.**

### Average Metrics in V1 template.

(A) Normalized average power for each stimulus. Gray lines are for each penetration, and red lines and shaded ribbons show Mean $\pm$ SEM. (B) Normalized average local coherence for each stimulus is shown similarly to (A). (C) Normalized average  $\Delta P/P$  in time window: [30, 100]ms after stimulus onset. (D) Normalized average  $\Delta CSD$  in the same time window of C. The color-mapped lines and shaded gray ribbons in C and D show Mean $\pm$ SEM. Square Markers are the same as in the figures of this article.

Figure S3.

### Coefficient of Variation.

Variability of normalized depth profiles (gray lines in [Figure S2](#)) across penetrations is accessed by coefficient of variation (STD/Mean) for power (red), local coherence (blue),  $\Delta P/P$  (orange), and  $\Delta CSD$  (green) in response to different stimulus conditions. Square Markers are the same as in the figures of this article.

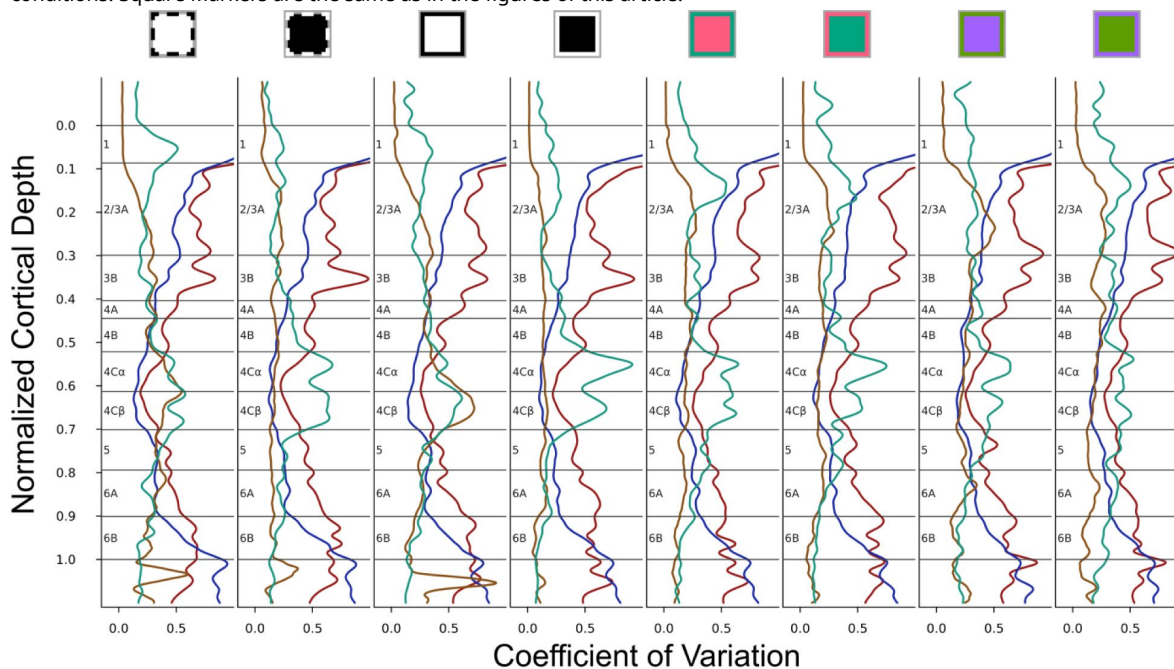
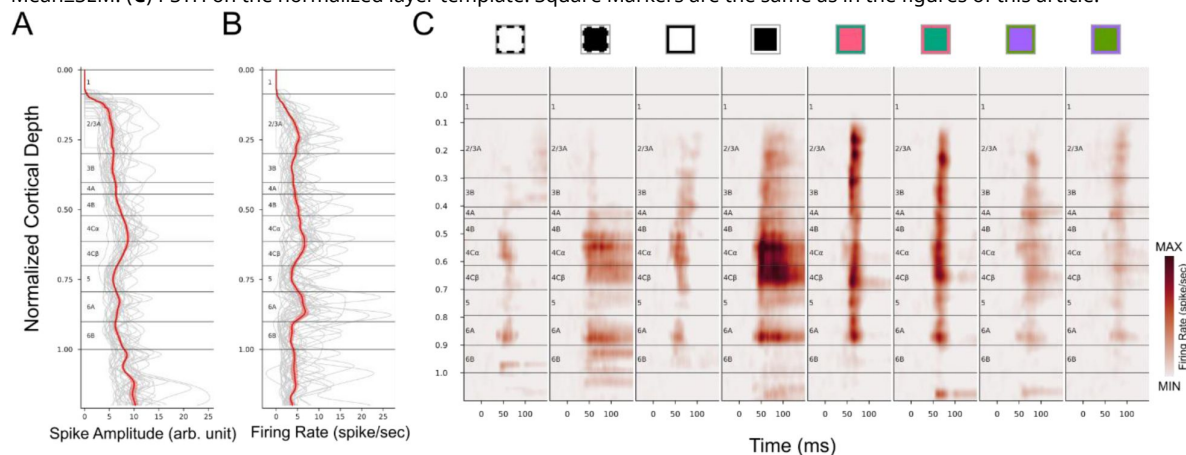


Figure S4.

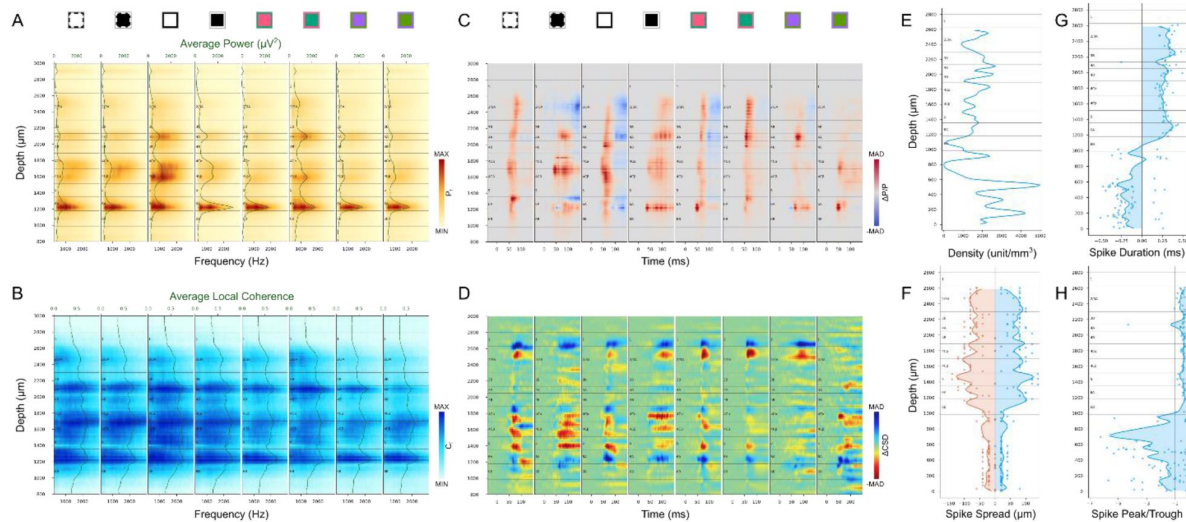
### Depth profile of spike amplitude and firing rate.

(A) Spike amplitude. (B) Firing rate during the whole recording session of a penetration. The stimulus set includes static/drift, achromatic/chromatic gratings. (A-B) gray lines for each penetration, red line and shaded ribbon for Mean $\pm$ SEM. (C) PSTH on the normalized layer template. Square Markers are the same as in the figures of this article.



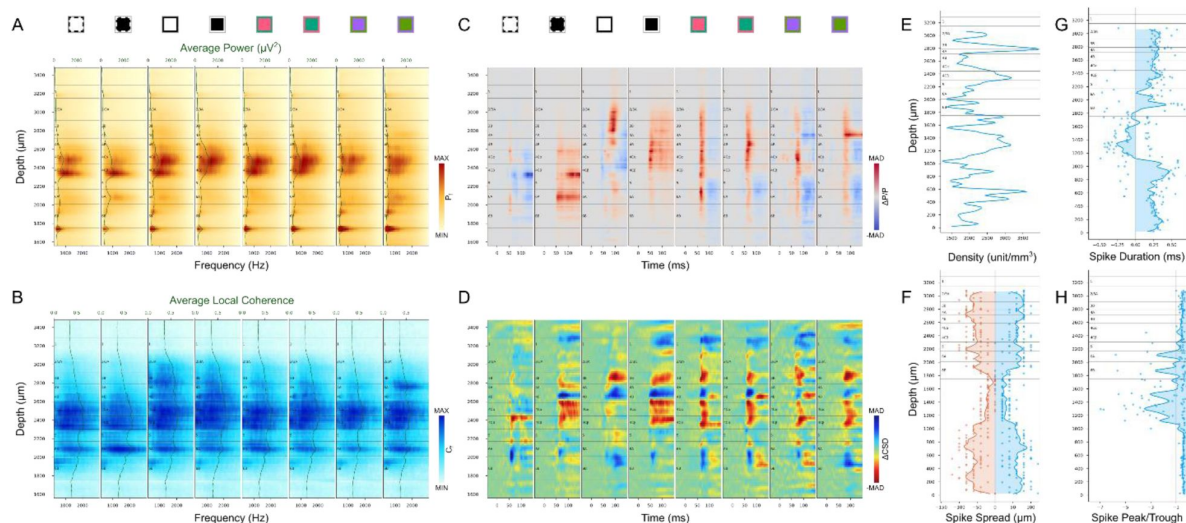
**Figure S5.**

**V1 layer delineation for example penetration (A3) based on power spectrum (A), local coherence (B), instantiations power changes (C), CSD changes (D), unit density (E), spike spread (F), spike duration (G) and spike peak-trough ratio (H).**



**Figure S6.**

**V1 layer delineation for example penetration (B3) based on power spectrum (A), local coherence (B), instantiations power changes (C), CSD changes (D), unit density (E), spike spread (F), spike duration (G) and spike peak-trough ratio (H).**

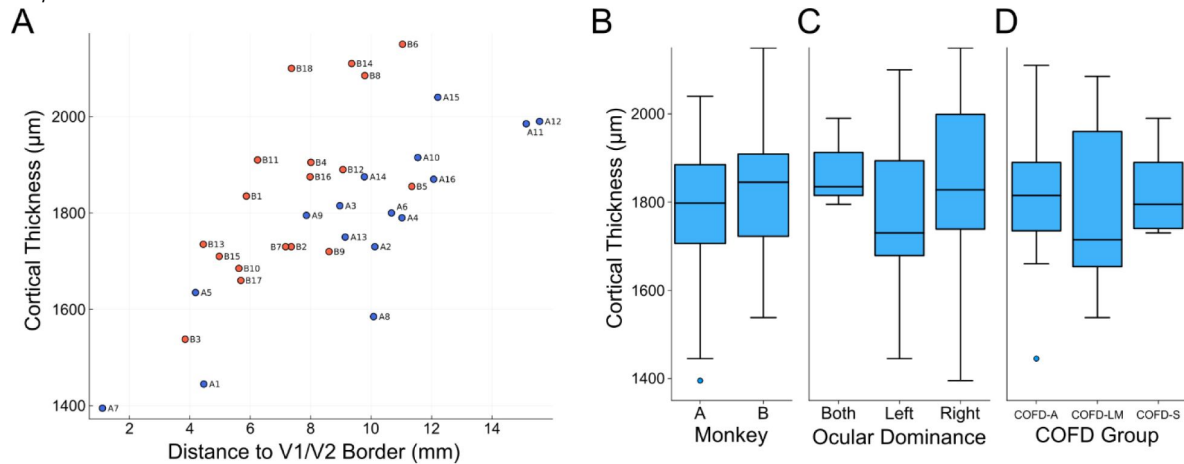




**Figure S7.**

### Thickness of primary visual cortex.

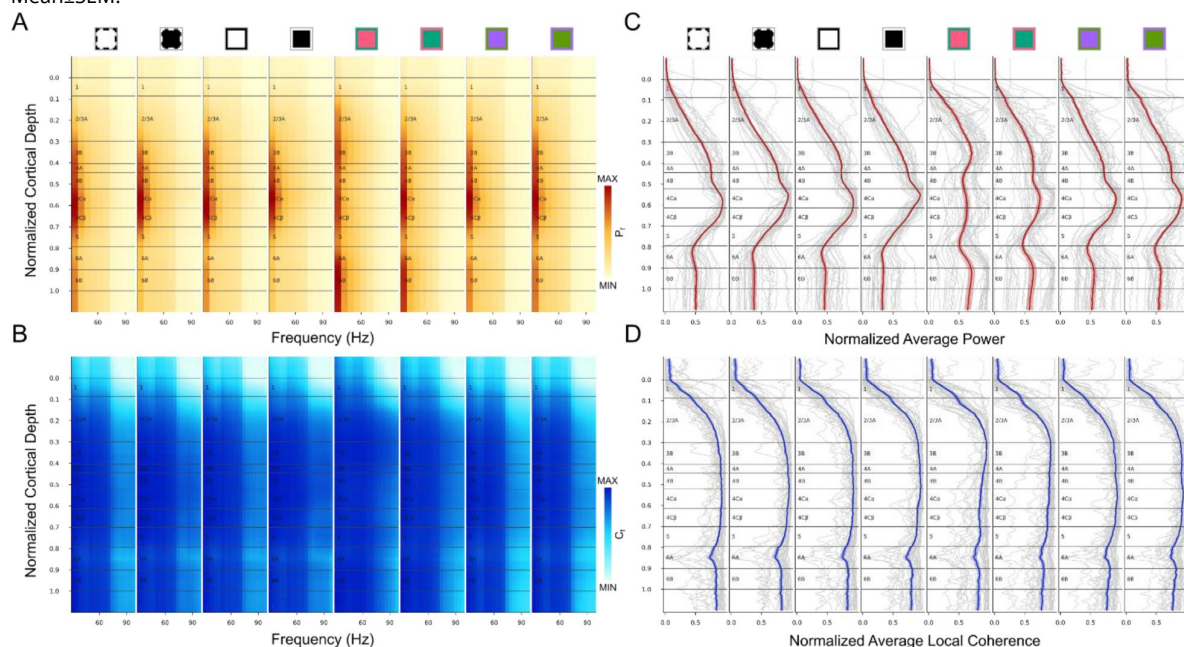
(A) The distance was measured from the location of penetration perpendicular to V1/V2 border. Here, all penetrations except A17 (penetration location information lost) were included. (B-D) Distributions of cortical thickness across two monkeys, different ocular dominance columns, and COFD groups. No significant differences were found in each category (Kruskal-Wallis Test,  $p > 0.05$ ). COFD-A: Achromatic ON/OFF domains; COFD-LM: L- and M-cone ON/OFF domains; COFD-S: S-cone ON/OFF domains.



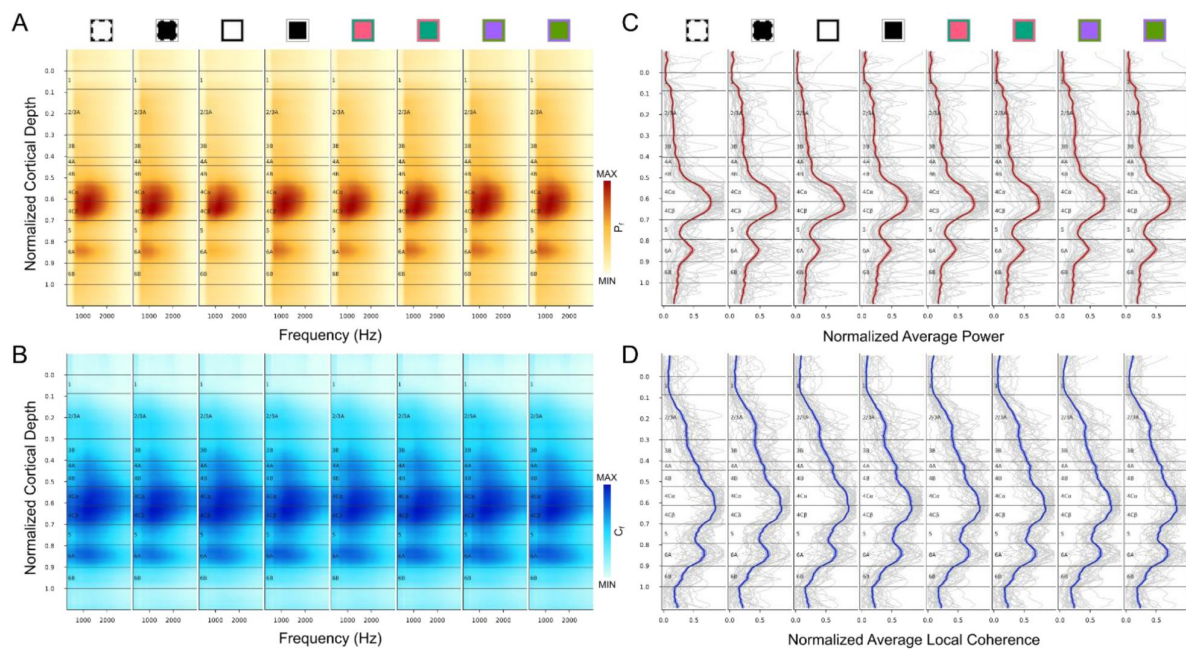
**Figure S8.**

### Average metrics from gamma band LFP in V1 template.

(A-B) Average gamma band (30-100Hz) spectrum profile of power (A) and local coherence (B) across penetrations. The square markers and color bars were the same as the figures in this article. (C-D) Normalized average power and local coherence in the gamma band. Gray lines are for each penetration, color-lines and corresponding shaded ribbons show Mean  $\pm$  SEM.



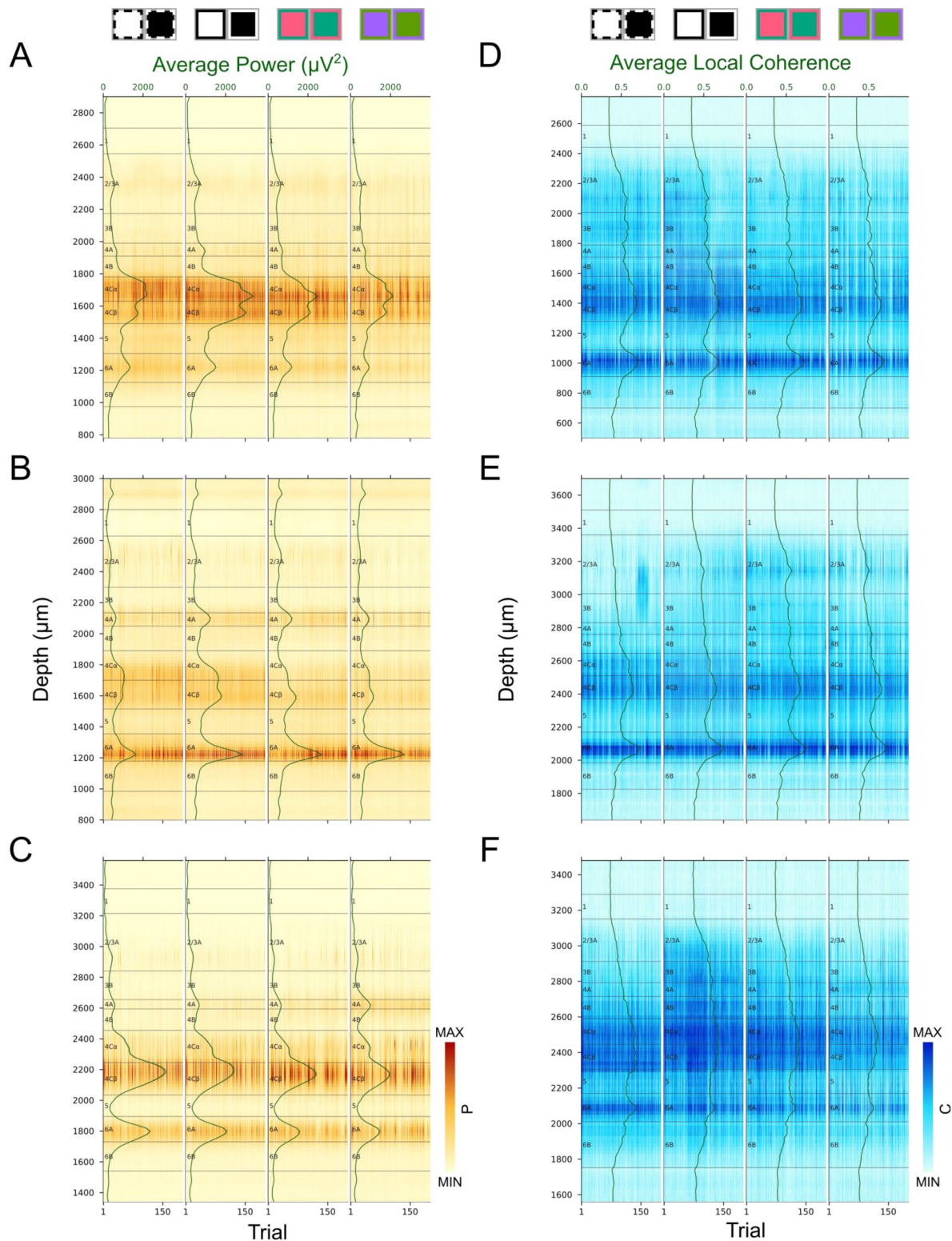




**Figure S9.**

### Average metrics from baseline window in V1 template.

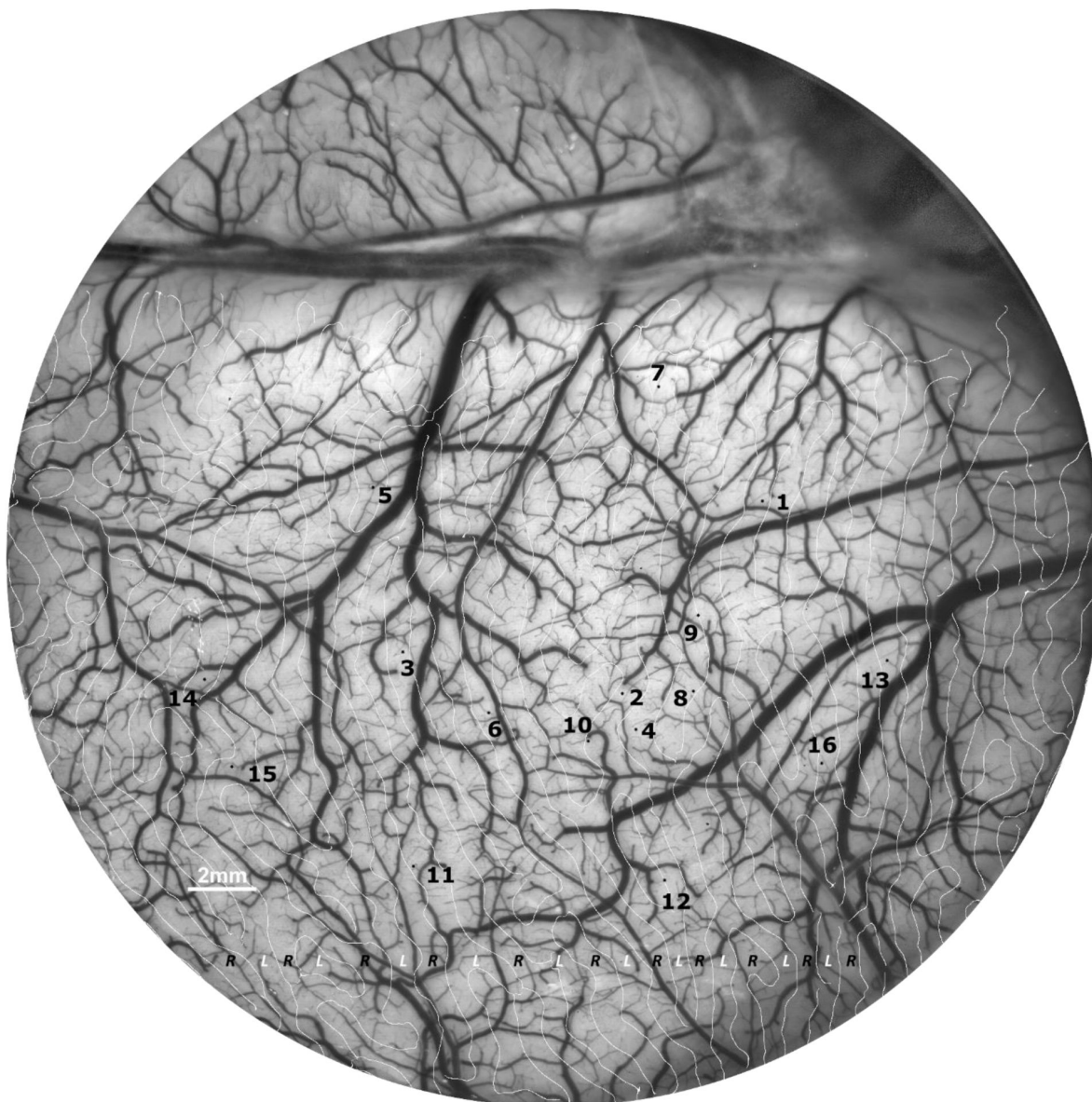
(A-B) Average baseline AP spectrum profile of power (A) and local coherence (B) across penetrations. The square markers and color bars were the same as the figures in this article. (C-D) Normalized average baseline AP power and local coherence. Gray lines are for each penetration, color-lines and corresponding shaded ribbons show Mean  $\pm$  SEM.



**Figure S10.**

**AP power and local coherence averaged across frequencies for each trial.**

(A-C) Powers for the same penetrations (A: A2, B: A3, C: B1) in [Figure 5](#). (D-F) Local coherences for the same penetrations (D: B12, E: B10, F: B3) in [Figure 6](#). Power and local coherence profiles were independently color-mapped according to the minimum and maximum in each profile. The green lines and corresponding shaded ribbons showed the Mean  $\pm$  SEM across trials.

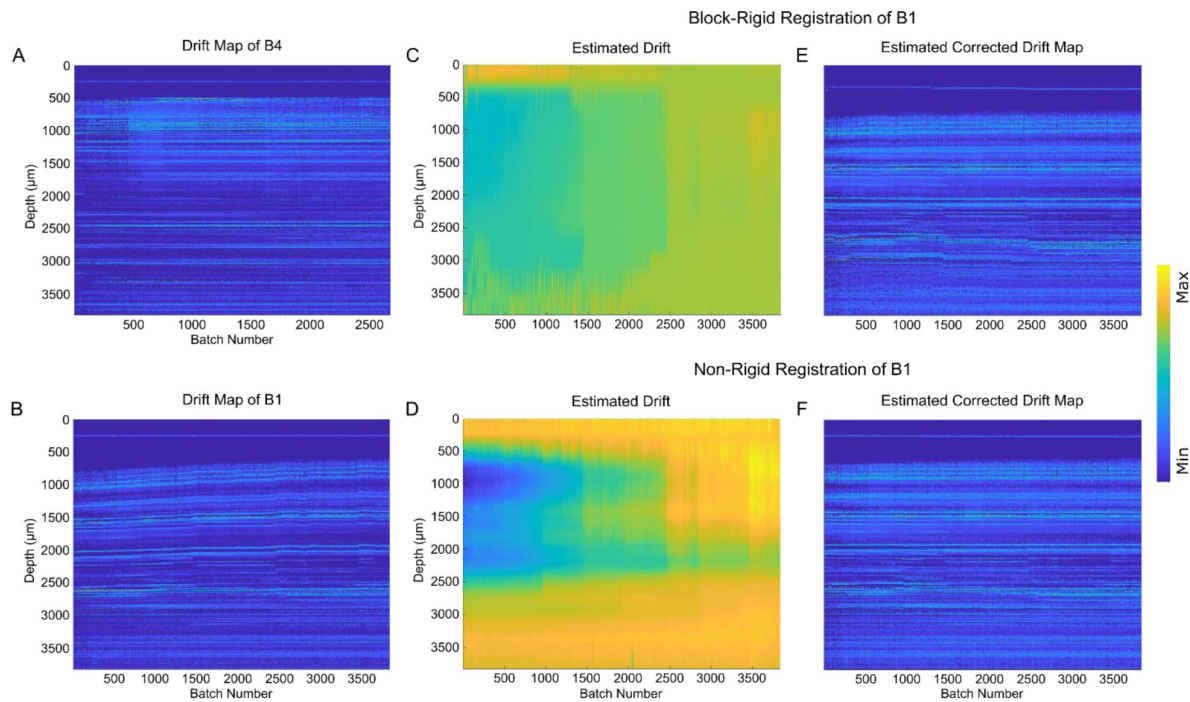


**Figure S11.**

**Penetration sites on V1 surface of monkey A.**

White lines together with L/R texts indicate the borders of ocular dominance column.





**Figure S12.**

### Electrodes drift correction.

(A) Probe drifting during the recording time for penetration B4 was visualized as average spike amplitudes shifting in consecutive batches (batch length:  $\sim 3.3$ sec). (B) Probe drifting of penetration B1 with same batch length as penetration B4. (C) Drift estimated for penetration B1 using the default method in Kilosort 3 (drift range:  $[-300, 300]$   $\mu\text{m}$ , number of blocks: 13, drift range for each block:  $[-300, 300]$   $\mu\text{m}$ ). (D) Drift estimated for penetration B1 using 'imregdemons' function in MATLAB. (E) Expected correction result of penetration B1 after applying ('imwarp' function in MATLAB) the estimated drift in C. (F) Expected correction result of penetration B1 after applying ('imwarp' function in MATLAB) the estimated drift in D.



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## Editors

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### Reviewer #1 (Public review):

Summary:

In this study, Zhang et al., presented an electrophysiology method to identify the layers of macaque visual cortex with high density Neuropixels 1.0 electrode. They found several electrophysiology signal profiles for high-resolution laminar discrimination and described a set of signal metrics for fine cortical layer identification.

Strengths:

There are two major strengths. One is the use of high density electrodes. The Neuropixels 1.0 probe has 20  $\mu\text{m}$  spacing electrodes, which can provide high resolution for cortical laminar identification. The second strength is the analysis. They found multiple electrophysiology signal profiles which can be used for laminar discrimination. Using this new method, they could identify the most thin layer in macaque V1. The data support their conclusion.

Weaknesses:

While this electrophysiology strategy is much easier to perform even in awake animals compared to histological staining methods, it provides an indirect estimation of cortical layers. A parallel histological study can provide a direct matching between the electrode signal features and cortical laminar locations. However, there are technical challenges, for example the distortions in both electrode penetration and tissue preparation may prevent a precise matching between electrode locations and cortical layers. In this case, additional micro wires electrodes binding with Neuropixels probe can be used to inject current and mark the locations of different depths in cortical tissue after recording.

<https://doi.org/10.7554/eLife.97290.2.sa3>

### Reviewer #2 (Public review):

Summary:

This paper documents a compelling attempt to accurately determine the locations and boundaries of the anatomically and functionally defined layers in macaque primary visual cortex using voltage signals recorded from a high-density electrode array that spans the full depth of cortex with contacts at 20  $\mu\text{m}$  spacing. First, the authors attempt to use current source density (CSD) analysis to determine layer locations, but they report a striking failure because the results vary greatly from one electrode penetration to the next and because the spatial resolution of the underlying local field potential (LFP) signal is coarse compared to the electrical contact spacing. The authors therefore turn to examining higher frequency signals related to action potentials and provide evidence that these signals reflect changes in neuronal size and packing density, response latency and visual selectivity, which taken



together can advance the state-of-the-art accuracy in making layer assignments from in vivo recordings.

#### Strengths:

There is a lot of nice data to look at in this paper that show interesting quantities as a function of depth in V1. Bringing all of these together offers the reader a rich data set: CSD, action potential shape, response power and coherence spectrum, and post-stimulus time response traces. Furthermore, data are displayed as a function of eye (dominant or non-dominant) and for achromatic and cone-isolating stimuli.

This paper takes a strong stand in pointing out weaknesses in the ability of CSD analysis to make consistent determinations about cortical layering in V1. Many researchers have found CSD to be problematic, and the observations here may be important to motivate other researchers to carry out rigorous comparisons and publish their results, even if they reflect negatively on the value of CSD analysis.

The paper provides a thoughtful, practical and comprehensive recipe for assigning traditional cortical layers based on easily-computed metrics from electrophysiological recordings in V1, and this is likely to be useful for electrophysiologists who are now more frequently using high-density electrode arrays.

#### Weaknesses:

Considerable space is taken in pointing out features that are well known, for example, the latency difference associated with different retinogeniculate pathways, the activity level differences associated with input layers, and the action potential shape differences associated with white vs. gray matter. These have been used for decades as indicators of depth and location of recordings in visual cortex as electrodes were carefully advanced. High density electrodes allow this type of data to now be collected in parallel, but at discrete, regular sampling points. Perhaps more emphasis could be placed on developing a rigorous analysis of how variable vs. reproducible are quantitative metrics of these features across penetrations, as a function of distance or functional domain, and from animal to animal, but this paper certainly makes a substantial push in this direction.

Another important piece of information for assessing the ability to determine layers from spiking activity is to carry out post-mortem histological processing so that the layer determination made in vivo can be compared to anatomical layering. However, histological methods also suffer from distortion and noise, thus it remains to be seen how much can ultimately be gained by integrating histology with the physiological methods explored here.

#### Overall

Overall, this paper makes a compelling argument in favor of using action potentials and stimulus driven responses, instead of CSD measurements, to assign cortical layers to electrode contacts in V1. The rich presentation of data, combined with the authors' highly educated interpretation and speculation about how useful such measurements will be for layer assignment make this an important paper for many labs using high-density electrodes. It is easy to agree with much of what is postulated here and to hope that we will soon have reliable, quantitative methods to make layer assignments that will be meaningful in terms of the differentiated roles of single neurons in cortical computation. How much this will end up corresponding to the canonical layer numbering that has been used for many decades will be interesting to see.

#### Comments on revisions:

I found that the authors addressed my main concerns to the degree they were able. They improved the consistency of language and figures, and they added some useful quantification.

<https://doi.org/10.7554/eLife.97290.2.sa2>

### Reviewer #3 (Public review):

#### Summary:

Zhang et al. explored strategies for aligning electrophysiological recordings from high-density laminar electrode arrays (Neuropixels) with the pattern of lamination across cortical depth in macaque primary visual cortex (V1), with the goal of improving the spatial resolution of layer identification based on electrophysiological signals alone. The authors compare the current commonly used standard in the field - current source density (CSD) analysis - with a new set of measures largely derived from action potential (AP) frequency band signals. Individual AP band measures provide distinct cues about different landmarks or potential laminar boundaries, and together they are used to subdivide the spatial extent of array recordings into discrete layers, including the very thin layer 4A, at a level of resolution unavailable when relying on CSD analysis alone for laminar identification. The authors compare the widths of the resulting subdivisions with previously reported anatomical measurements as evidence that layers have been accurately identified. This is a bit circular, given that they also use these anatomical measurements as guidelines limiting the boundary assignments; however, the strategy is overall sensible and the electrophysiological signatures used to identify layers are generally convincing. Furthermore, by varying the pattern of visual stimulation to target chromatically sensitive inputs known to be partially segregated by layer in V1, they show localized response patterns that lend confidence to their identification of particular sublayers.

The authors compellingly demonstrate the insufficiency of CSD analysis for precisely identifying fine laminar structure, and in some cases its limited accuracy at identifying coarse structure. CSD analysis produced inconsistent results across array penetrations and across visual stimulus conditions and was not improved in spatial resolution by sampling at high density with Neuropixels probes. Instead, in order to generate a typical, informative pattern of current sources and sinks across layers, the LFP signals from the Neuropixels arrays required spatial smoothing or subsampling to approximately match the coarser (50-100  $\mu\text{m}$ ) spacing of other laminar arrays. Even with smoothing, the resulting CSDs in some cases predicted laminar boundaries that were inconsistent with boundaries estimated using other measures and/or unlikely given the typical sizes of individual layers in macaque V1. This point alone provides an important insight for others seeking to link their own laminar array recordings to cortical layers.

They next offer a set of measures based on analysis of AP band signals. These measures include analyses of the density, average signal spread, and spike waveforms of units identified through spike sorting, as well as analyses of AP band power spectra and local coherence profiles across recording depth. The power spectrum measures in particular yield compact peaks at particular depths, albeit with some variation across penetrations, whereas the waveform measures most convincingly identified the layer 6-white matter transition. In general, some of the new measures yield inconsistent patterns across penetrations, and some of the authors' explanations of these analyses draw intriguing but rather speculative connections to properties of anatomy and/or responsivity. However, taken as a group, the set of AP band analyses appear sufficient to determine the layer 6-white matter transition with precision and to delineate intermediate transition points likely to correspond to actual layer

boundaries, and the strategy serves as a substantial advancement over consideration of CSD signals alone to match electrophysiological recordings with cortical layers.

#### Strengths:

The authors convincingly demonstrate the potential to resolve putative laminar boundaries using only electrophysiological recordings from Neuropixels arrays. This is particularly useful given that histological information is often unavailable for chronic recordings. They make a clear case that CSD analysis is insufficient to resolve the lamination pattern with the desired precision and offer a thoughtful set of alternative analyses, along with an order in which to consider multiple cues in order to facilitate others' adoption of the strategy. The suggested analyses can be used to reliably identify certain landmarks (the positions of layer 4c and the layer 6-white matter transition), which provide very useful constraints for specifying the remaining laminar boundaries, and consideration of average anatomical patterns makes it unlikely that the remaining laminar boundaries will be far from their true locations. Overall, the widths of the resulting layers bear a sensible resemblance to the expected widths identified by prior anatomical measurements, and at least in some cases there are satisfying signatures of chromatic visual sensitivity and latency differences across layers that are predicted by the known connectivity of the corresponding layers. Thus, the proposed analytical toolkit appears to work well for macaque V1 and has strong potential to generalize to use in other cortical regions, though area-targeted selection of stimuli may be required.

#### Weaknesses:

The waveform measures, in particular the unit density distribution, are likely to be sensitive to the methods and criteria used for spike sorting, which differ among experimenters/groups, and this may limit the usefulness of this particular measure for others in the community. More generally, although the sizes of identified layers comport with typical sizes identified anatomically, a more powerful confirmation would be a direct comparison with histologically identified boundaries along each penetration's trajectory. Ultimately, the absence of this type of independent confirmation limits the strength of the claim that veridical laminar boundaries can be precisely identified from electrophysiological signals alone.

<https://doi.org/10.7554/eLife.97290.2.sa1>

#### Author response:

The following is the authors' response to the original reviews.

##### **Public Reviews:**

##### **Reviewer #1 (Public Review):**

##### *Summary:*

*In this study, Zhang et al., presented an electrophysiology method to identify the layers of macaque visual cortex with high density Neuropixels 1.0 electrode. They found several electrophysiology signal profiles for high-resolution laminar discrimination and described a set of signal metrics for fine cortical layer identification.*

##### *Strengths:*

*There are two major strengths. One is the use of high density electrodes. The Neuropixels 1.0 probe has 20 um spacing electrodes, which can provide high resolution for cortical laminar identification. The second strength is the analysis. They found multiple*

*electrophysiology signal profiles which can be used for laminar discrimination. Using this new method, they could identify the most thin layer in macaque V1. The data support their conclusion.*

**Weaknesses:**

*While this electrophysiology strategy is much easier to perform even in awake animals compared to histological staining methods, it provides an indirect estimation of cortical layers. A parallel histological study can provide a direct matching between the electrode signal features and cortical laminar locations. However, there are technical challenges, for example the distortions in both electrode penetration and tissue preparation may prevent a precise matching between electrode locations and cortical layers. In this case, additional micro wires electrodes binding with Neuropixels probe can be used to inject current and mark the locations of different depths in cortical tissue after recording.*

While we agree that it would be helpful to adopt a more direct method for linking laminar changes observed with electrophysiology to anatomical layers observed in postmortem histology, we do not believe that the approach suggested by the reviewer would be particularly helpful. The approach suggested involves making lesions, which are known to be quite variable in size, asymmetric in shape, and do not have a predictable geometry relative to the location of the electrode tip. In contrast, our electrophysiology measures have identified clear boundaries which precisely match the known widths and relative positions of all the layers of V1, including layer 4A, which is only 50 microns thick, much smaller than the resolution of lesion methods.

**Reviewer #2 (Public Review):**

**Summary:**

*This paper documents an attempt to accurately determine the locations and boundaries of the anatomically and functionally defined layers in macaque primary visual cortex using voltage signals recorded from a high-density electrode array that spans the full depth of cortex with contacts at 20  $\mu$ m spacing. First, the authors attempt to use current source density (CSD) analysis to determine layer locations, but they report a striking failure because the results vary greatly from one electrode penetration to the next and because the spatial resolution of the underlying local field potential (LFP) signal is coarse compared to the electrical contact spacing. The authors thus turn to examining higher frequency signals related to action potentials and provide evidence that these signals reflect changes in neuronal size and packing density, response latency and visual selectivity.*

**Strengths:**

*There is a lot of nice data to look at in this paper that shows interesting quantities as a function of depth in V1. Bringing all of these together offers the reader a rich data set: CSD, action potential shape, response power and coherence spectrum, and post-stimulus time response traces. Furthermore, data are displayed as a function of eye (dominant or non-dominant) and for achromatic and cone-isolating stimuli.*

*This paper takes a strong stand in pointing out weaknesses in the ability of CSD analysis to make consistent determinations about cortical layering in V1. Many researchers have found CSD to be problematic, and the observations here may be important to motivate other researchers to carry out rigorous comparisons and publish their results, even if they reflect negatively on the value of CSD analysis.*

*The paper provides a thoughtful, practical and comprehensive recipe for assigning traditional cortical layers based on easily-computed metrics from electrophysiological*



*recordings in V1, and this is likely to be useful for electrophysiologists who are now more frequently using high-density electrode arrays.*

*Weaknesses:*

*Much effort is spent pointing out features that are well known, for example, the latency difference associated with different retinogeniculate pathways, the activity level differences associated with input layers, and the action potential shape differences associated with white vs. gray matter. These have been used for decades as indicators of depth and location of recordings in visual cortex as electrodes were carefully advanced. High density electrodes allow this type of data to now be collected in parallel, but at discrete, regular sampling points. Rather than showing examples of what is already accepted, the emphasis should be placed on developing a rigorous analysis of how variable vs. reproducible are quantitative metrics of these features across penetrations, as a function of distance or functional domain, and from animal to animal. Ultimately, a more quantitative approach to the question of consistency is needed to assess the value of the methods proposed here.*

We thank the reviewer for suggesting the addition of quantitative metrics to allow more substantive comparisons between various measures within and between penetrations. We have added quantification and describe this in the context of more specific comments made by this reviewer. We have retained descriptions of metrics that are well established because they provide an important validation of our approaches and laminar assignments.

*Another important piece of information for assessing the ability to determine layers from spiking activity is to carry out post-mortem histological processing so that the layer determination made in this paper could be compared to anatomical layering.*

We are not aware of any approach that would provide such information at sufficient resolution. For example, it is well known that electrolytic lesions often do not match to the locations expected from electrophysiological changes observed with single electrodes. As noted above, our observation that the laminar changes in electrophysiology precisely match the known widths and relative positions of all the layers of V1, including layer 4A, provides confidence in our laminar assignments.

*On line 162, the text states that there is a clear lack of consistency across penetrations, but why should there be consistency: how far apart in the cortex were the penetrations? How long were the electrodes allowed to settle before recording, how much damage was done to tissue during insertion? Do you have data taken over time - how consistent is the pattern across several hours, and how long was the time between the collection of the penetrations shown here?*

Answers to most of these questions can be found within the manuscript text. We have added text describing distance between electrode penetrations (at least 1mm, typically far more) and added a figure which shows a map of the penetration locations. The Methods section describes electrode penetration methods to minimize damage and settling times of penetrations. Data are provided regarding changes in recordings over time (see Methods, Drift Correction). The stimuli used to generate the data described are presented within a total of 30 minutes or less, minimizing any changes that might occur due to electrode drift. There is a minimum of 3 hours between different penetrations from the same animal.

*The impact of the paper is lessened because it emphasizes consistency but not in a consistent manner. Some demonstrations of consistency are shown for CSDs, but not quantified. Figure 4A is used to make a point about consistency in cell density, but across animals, whereas the previous text was pointing out inconsistency across penetrations.*

*What if you took a 40 or 60 um column of tissue and computed cell density, then you would be comparing consistency across potentially similar scales. Overall, it is not clear how all of these different metrics compare quantitatively to each other in terms of consistency.*

As noted above, we have now added quantitative comparisons of consistency between different metrics. It is unclear why the reviewer felt that we use Figure 4A to describe consistency. That figure was a photograph from a previous publication simply showing the known differences in neuron density that are used to define layers in anatomical studies. This was intended to introduce the reader to known laminar differences. At any rate, we have been unable to contact the previous publishers of that work to obtain permission to use the figure. So we have removed that figure as it is unnecessary to illustrate the known differences in cell density that are used to define layers. We have kept the citation so that interested readers can refer to the publication.

*In many places, the text makes assertions that A is a consistent indicator of B, but then there appear to be clear counterexamples in the data shown in the figures. There is some sense that the reasoning is relying too much on examples, and not enough on statistical quantities.*

Without reference to specific examples we are not able to address this point.

#### *Overall*

*Overall, this paper makes a solid argument in favor of using action potentials and stimulus driven responses, instead of CSD measurements, to assign cortical layers to electrode contacts in V1. It is nice to look at the data in this paper and to read the authors' highly educated interpretation and speculation about how useful such measurements will be in general to make layer assignments. It is easy to agree with much of what they say, and to hope that in the future there will be reliable, quantitative methods to make meaningful segmentations of neurons in terms of their differentiated roles in cortical computation. How much this will end up corresponding to the canonical layer numbering that has been used for many decades now remains unclear.*

#### **Reviewer #3 (Public Review):**

##### *Summary:*

*Zhang et al. explored strategies for aligning electrophysiological recordings from high-density laminar electrode arrays (Neuropixels) with the pattern of lamination across cortical depth in macaque primary visual cortex (V1), with the goal of improving the spatial resolution of layer identification based on electrophysiological signals alone. The authors compare the current commonly used standard in the field - current source density (CSD) analysis - with a new set of measures largely derived from action potential (AP) frequency band signals. Individual AP band measures provide distinct cues about different landmarks or potential laminar boundaries, and together they are used to subdivide the spatial extent of array recordings into discrete layers, including the very thin layer 4A, a level of resolution unavailable when relying on CSD analysis alone for laminar identification. The authors compare the widths of the resulting subdivisions with previously reported anatomical measurements as evidence that layers have been accurately identified. This is a bit circular, given that they also use these anatomical measurements as guidelines limiting the boundary assignments; however, the strategy is overall sensible and the electrophysiological signatures used to identify layers are generally convincing. Furthermore, by varying the pattern of visual stimulation to target chromatically sensitive inputs known to be partially segregated by layer in V1, they show*

localized response patterns that lend confidence to their identification of particular sublayers.

*The authors compellingly demonstrate the insufficiency of CSD analysis for precisely identifying fine laminar structure, and in some cases its limited accuracy at identifying coarse structure. CSD analysis produced inconsistent results across array penetrations and across visual stimulus conditions and was not improved in spatial resolution by sampling at high density with Neuropixels probes. Instead, in order to generate a typical, informative pattern of current sources and sinks across layers, the LFP signals from the Neuropixels arrays required spatial smoothing or subsampling to approximately match the coarser (50-100  $\mu$ m) spacing of other laminar arrays. Even with smoothing, the resulting CSDs in some cases predicted laminar boundaries that were inconsistent with boundaries estimated using other measures and/or unlikely given the typical sizes of individual layers in macaque V1. This point alone provides an important insight for others seeking to link their own laminar array recordings to cortical layers.*

*They next offer a set of measures based on analysis of AP band signals. These measures include analyses of the density, average signal spread, and spike waveforms of single- and multi-units identified through spike sorting, as well as analyses of AP band power spectra and local coherence profiles across recording depth. The power spectrum measures in particular yield compact peaks at particular depths, albeit with some variation across penetrations, whereas the waveform measures most convincingly identified the layer 6-white matter transition. In general, some of the new measures yield inconsistent patterns across penetrations, and some of the authors' explanations of these analyses draw intriguing but rather speculative connections to properties of anatomy and/or responsivity. However, taken as a group, the set of AP band analyses appear sufficient to determine the layer 6-white matter transition with precision and to delineate intermediate transition points likely to correspond to actual layer boundaries.*

#### **Strengths:**

*The authors convincingly demonstrate the potential to resolve putative laminar boundaries using only electrophysiological recordings from Neuropixels arrays. This is particularly useful given that histological information is often unavailable for chronic recordings. They make a clear case that CSD analysis is insufficient to resolve the lamination pattern with the desired precision and offer a thoughtful set of alternative analyses, along with an order in which to consider multiple cues in order to facilitate others' adoption of the strategy. The widths of the resulting layers bear a sensible resemblance to the expected widths identified by prior anatomical measurements, and at least in some cases there are satisfying signatures of chromatic visual sensitivity and latency differences across layers that are predicted by the known connectivity of the corresponding layers. Thus, the proposed analytical toolkit appears to work well for macaque V1 and has strong potential to generalize to use in other cortical regions, though area-targeted selection of stimuli may be required.*

#### **Weaknesses:**

*The waveform measures, and in particular the unit density distribution, are likely to be sensitive to the criteria used for spike sorting, which differ widely among experimenters/groups, and this may limit the usefulness of this particular measure for others in the community. The analysis of detected unit density yields fluctuations across cortical depth which the authors attribute to variations in neural density across layers; however, these patterns seemed particularly variable across penetrations and did not consistently yield peaks at depths that should have high neuronal density, such as layer 2. Therefore, this measure has limited interpretability.*

While we agree that our electrophysiological measure of unit density does not strictly reflect anatomical neuronal density, we would like to remind the reader that we use this measure only to roughly estimate the correspondence between changes in density and likely layer assignments. We rely on other measures (e.g. AP power, AP power changes in response to visual stimuli) that have sharp borders and more clear transitions to assign laminar boundaries. Further, as noted in the reviewer's list of strengths, the laminar assignments made with these measures are cross validated by differences in response latencies and sensitivity to different types of stimuli that are observed at different electrode depths.

*More generally, although the sizes of identified layers comport with typical sizes identified anatomically, a more powerful confirmation would be a direct per-penetration comparison with histologically identified boundaries. Ultimately, the absence of this type of independent confirmation limits the strength of their claim that veridical laminar boundaries can be identified from electrophysiological signals alone.*

As we have noted in response to similar comments from other reviewers, we are not aware of a method that would make this possible with sufficient resolution.

#### **Recommendations for the authors:**

##### **Reviewing Editor (Recommendations For The Authors):**

*The reviewers have indicated that their assessment would potentially be stronger if their advice for quantitative, statistically validated comparisons was followed, for example, to demonstrate variability or consistency of certain measures that are currently only asserted. Also, if available, some histological confirmation would be beneficial. It was requested that the use and modification of the layering from Balaram & Kaas is addressed, as well as dealing with inconsistencies in the scale bars on those figures. There are two figure permission issues that need to be resolved prior to publication: Balaram & Kaas 2014 in Fig 1A, Kelly & Hawken 2017 in Fig. 4A.*

Please see detailed responses to reviewer comments below. We have added new supplemental figures to quantitatively compare variability among metrics. As noted above, the suggested addition of data linking the electrophysiology directly to anatomical observations of laminar borders from the same electrode penetration is not feasible. The figure reused in Figure 1A is from open-access (CC BY) publication (Balaram & Kaas 2014). After reexamining the figure in the original study, we found that the inferred scale bar would give an obviously inaccurate result. So, we decided to remove the scale bar in Figure 1A. We haven't received any reply from Springer Nature for Figure 4A permission, so we decided to remove the reused figure from our article (Kelly & Hawken 2017).

##### **Reviewer #1 (Recommendations For The Authors):**

*Figure 4A has a different scale to Figure 4B-4F. It is better to add dashed lines to indicate the relationship between the cortical layers or overall range from Figure 4A to the corresponding layers in 4B to 4F.*

The reused figure in Figure 4A is removed due to permission issue. See also comments above.

##### **Reviewer #2 (Recommendations For The Authors):**

###### *General comments*

*This paper demonstrates that voltage signals in frequency bands higher than those used for LFP/CSD analysis can be used from high-density electrical contact recording to*



*generate a map of cortical layering in macaque V1 at a higher spatial resolution than previously attained.*

*My main concern is that much of this paper seems to show that properties of voltage signals recorded by electrodes change with depth in V1. This of course is well known and has been mapped by many who have advanced a single electrode micron-by-micron through the cortex, listening and recording as they go. Figure 4 shows that spike shapes can give a clear indication of GM to WM borders, and this is certainly true and well known. Figures 5 and 6 show that activity level on electrodes can indicate layers related to LGN input, and this is known. Figure 7 shows that latencies vary with layer, and this is certainly true as we know. A main point seems to be that CSD is highly inconsistent. This is important to know if CSD is simply never going to be a good measure for layering in V1, but it would require quantification and statistics to make a fair comparison.*

We are glad to see that the reviewer understands that changes in electrical signals across layers are well known and are expected to have particular traits that change across layers. We do not claim that have discovered anything that is unexpected or unknown. Instead, we introduce quantitative measures that are sensitive to these known differences (historically, often just heard with an audio monitor e.g. “LGN axon hash”). While the primary aim of this paper is not to show that Neuropixels probes can record some voltage signal properties that cannot be recorded with a single electrode before, we would like to point out that multi-electrode arrays have a very different sampling bias and also allow comparisons of simultaneous recordings across contacts with known fixed distances between them. For example our measure of “unit spread” could not be estimated with a single electrode.

We’ve added Figure S3 to show quantitative comparison of variation between CSD and AP metrics. These figures add support to our prior, more anecdotal descriptions showing that CSDs are inconsistent and lack the resolution needed to identify thin layers.

*Some things are not explained very clearly. Like achromatic regions, and eye dominance - these are not quantified, and we don't know if they are mutually consistent - are achromatic/chromatic the same when tested through separate eyes? How consistent are these basic definitions? How definitive are they?*

The quantitative definitions of achromatic region/COFD and eye dominance column can be found in our previous paper (Li et al., 2022) cited in this article. The main theme of this study is to develop a strategy for accurately identifying layers, the more detailed functional analysis will be described in future publications.

#### *Specific comments*

*The abstract refers to CSD analysis and CSD signals. Can you be more precise - do you aim to say that LFP signals in certain frequency bands are already known to lack spatial localization, or are you claiming to be showing that LFP signals lack spatial resolution? A major point of the results appears to be lack of consistency of CSD, but I do not see that in the Abstract. The first sentence in the abstract appears to be questionable based on the results shown here for V1.*

We have updated the Abstract to minimize confusion and misunderstanding.

*Scale bar on Fig 1A implies that layers 2-5 are nearly 3 mm thick. Can you explain this thickness? Other figures here suggest layers 1-6 is less than 2 mm thick. Note, in a paper by the same authors (Balaram et al) the scale bar (100 um, Figure 4) on similar macaque tissue suggests that the cortex is much thinner than this. Perhaps neither is correct, but you should attempt to determine an approximately accurate scale. The text defines*

*granular as Layer 4, but the scale bar in A implies layer 4 is 1 mm thick, but this does not match the ~0.5 mm thickness consistent with Figure 1E, F. The text states that L4A is less than 100  $\mu$ m thick, but the markings and scale bar in Figure 1A suggests that it could be more than 100  $\mu$ m thick.*

We thank the reviewer for pointing out that there are clearly errors in the scale bars used in these previously published figures from another group. In the original figure 1 (Balaram & Kaas 2014), histological slices were all scaled to one of the samples (Chimpanzee) without scale bar. After reexamining the scale bar we derived based on figure 2 of the original study, we found the same problem. Since relative widths of layers are more important than absolute widths in our study, we decided to remove the scale bar that we had derived and added to the Figure 1A.

*Line 157. Fix "The most commonly visual stimulus"*

Text has been changed

*Line 161. Fix "through dominate eye"*

Text has been changed

*Line 166. Please specify if the methods established and validated below are histological, or tell something about their nature here.*

The Abstract and Introduction already described the nature of our methods

*Line 184. Text is mixing 'dominant' and 'dominate', the former is better.*

Text has been changed accordingly

*Line 188. Can you clarify "beyond the time before a new stimulus transition". Are you generally referring to the fact that neuronal responses outlast the time between changes in the stimulus?*

That is correct. We are referring to the fact that neuronal responses outlast the time between changes in the stimulus. We have edited the text for clarity.

*Line 196. Fix "dominate eye" in two places.*

Text has been changed

*Line 196. The text seems to imply it is striking to find different response patterns for the two eyes, but given the OD columns, why should this be surprising?*

Since we didn't find systematic comparison for CSD depth profiles of dominant/non-dominant eyes, or black/white in the past studies, we just describe what we saw in our data. The rationale for testing each eye is that it is known that LGN projections from two eyes remain separated in direct input layer of V1, so comparing CSDs from two eyes could potentially help identifying input layers, such as L4C. Here we provide evidence showing that CSD profiles from two eyes deviate from naive expectations. For example, CSDs from black stimulus show less variation between two eyes, whereas CSDs from white stimulus could range from similar profile to drastically different ones across eyes.

*Line 198. Text like, "The most consistent..." is stating overall conclusions drawn by the authors before pointing the reader specifically to the evidence or the quantification that supports the statement.*

We've adjusted the text pointing to Figure S2, where depth profiles of all penetrations are visualized, and a newly added Figure S3, where the coefficients of variation for several metric profiles were shown.

*Line 200. "white stimulus is more variable" - the text does not tell us where/how this is supported with quantitative analysis/statistics.*

We've adjusted the text pointing to Figure S2, S3

*The metric in 4B is not explained, the text mentions the plot but the reader is unable to make any judgement without knowledge of the method, nor any estimate of error bars.*

The figure is first mentioned in section: Unit Density, and text in this section already described the definition of neuron density and unit density. We've also modified the text pointing to the method section for details.

*Line 236. The text states the peak corresponds to L4C, but does not explain how the layer lines were determined.*

As described early in the CSD section, all layer boundaries are determined following the guide which layouts the strategy for how to draw borders by combining all metrics.

*At Line 296 the spike metrics section ends without providing a clear quantification of how useful the metrics will be. It is clear that the GM to WM boundary can be identified, but that can be found with single electrodes as well, as neurophysiologists get to see/hear the change in waveform as the electrode is advanced in even finer spatial increments than the 20 um spacing of the contacts here.*

The aim of this study is to develop an approach for accurately delineating layers simultaneously. The metrics we explored are considered estimation of well-known properties, so they can provide support for the correctness we hope to achieve. Here we first demonstrate the usefulness and later show the average across penetrations (Figure 9C-F). We are less concerned in quantification of how different factors affect precision and consistency of these metrics or how useful a single metric is, but rather, as described in the guide section, whether we can delineate all layers given all metrics.

*Line 302-306. Why this statement is made here is unclear, it interrupts the flow for a reason that perhaps will be explained later.*

This statement notes the insensitivity of this measure to temporal differences, introducing the value of incorporating a measure of how AP powers changes over time in the next section of the manuscript.

*Line 311. What is the reason to speculate about no canceling because of temporal overlap? Are you assuming a very sparse multi unit firing rate such that collisions do not happen?*

Here we describe a simple theoretical model in which spike waveforms only add without cancelling, then the power would be proportional to the number of spikes. In reality, spike

waveform sometimes cancels causing the theoretical relationship to deteriorate to some degree.

*Lines 327-346. There is a considerable amount of speculation and arguing based on particular examples and there is a lack of quantification. Neuron density is mentioned, but not firing rate. would responses from fewer neurons with higher firing rate not be similar to more neurons with lower firing rates?*

According to the theoretical model we described, power is proportional to numbers of spikes which then depend on both neuron density and firing rate. So fewer neurons with higher firing rate would generate similar power to more neurons with lower firing rate. We've expanded the explanation of the model and added Figure S4 about the depth profile of firing rate. Text has also been adjusted pointing to the Figure S2, S3 about quantitatively comparisons of variability.

*Line 348 states there is a precise link between properties and cortical layers, but the manuscript has not, up to this point, shown how that link was determined or quantified it.*

Through our generative model of power and the similarity between depth profile of firing rate and depth profile of neuron density (Figure S4), depth profile of power can be used to approximate depth profile of neuron density which is known to be closely correlated to cortical layering.

*Line 350. What is meant by "stochastic variability"?*

The text essentially says distances from electrode contact to nearby cell bodies were random, so closer cells have higher spike amplitudes and in turn result in higher power on a channel.

*The figures showing the two metrics, Pf and Cf, should be shown for the same data sets. The markings indicate that Fig 5 and Fig 6 show results from non-overlapping data sets. This does not build confidence about the results in the paper.*

Here we use typical profiles to demonstrate the characteristics of the power spectrum/coherence spectrum because of the variation across penetrations. We show later, in the guide section, all metrics for one penetration (another two cases in supplemental figures) and how to combine all metrics to derive layer delineations.

*Line 375 the statement is somewhat vague, "there are nevertheless sometimes cases where they can resolve uncertainties," can you please provide some quantitative support?*

We provided 3 examples in Figure 6, and more examples are shown in Figure 8, Figure S5, S6.

*Line 379. I believe the change you want to describe here is a change associated with a transition in the visual stimulus. It would be good to clarify this in the first several sentences here. Baseline can mean different things. I got the impression that your stimuli flip between states at a rate fast enough that signals do not really have time to return to a baseline.*

We rephrased the sentence to describe the metric more precisely. A pair of uniform colors flipping in 1.5 second intervals is usually long enough for spiking activities to decay to a saturated level.

*This section (379 - 398) continues a qualitative show-and-tell feel. There appears to be a lot of variability across the examples in Figure 7. How could you try to quantify this variability versus the variability in LFP? And, in this section overall, the text and figure legend don't really describe what the baseline is.*

Text adjustments are made to briefly describe the baseline window and point to the Method section where definitions are described in detail. We've added Figure S3 together with Figure S2 to address the variability across penetrations, stimuli, and metrics.

*Line 405 - 415. The discussion here does not consider that layers may not have well defined boundaries, the text gives the impression that there is some ultimate ground truth to which the metrics are being compared, but that may not be accurate.*

Except for a few layers/sublayers, such as L2, L3A, L3B, most layer boundaries of neocortex are well defined (Figure 1A) and histological staining of neurons/density and correlated changes in chemical content show very sharp transitions. The best of these staining methods is cytochrome oxidase, which shows sharp borders at the top and bottom of layer 4A, top and bottom of layer 4C, and the layer 5/6 border. There is also a sharp transition in neuronal cell body size and density at the top and bottom of layer 4Cb. The definition and delineation of all possible layers are constantly being refined, especially by accumulated knowledge of genetic markers of different cell types and connection patterns. In our study, we develop metrics to estimate well known anatomical and functional properties of different layers. We have also discussed layer boundaries that have been ambiguous to date and explained the reason and criteria to resolve them.

*Line 423. The text references Figure 1A in stating that relative thickness and position is crucial, but Figure 1A does not provide that information and does not explain how it might be determined, or how much of a consensus there is. Also, the text does not consider that the electrode may go through the cortex at oblique angles, and not the same angle in each layer, and the relative thickness may not be a dependable reference.*

There are numerous studies that describe criteria to delineate cortical layers, the referenced article (Balaram & Kaas 2014) is used here as an example. We are not aware of any publication that has systematically compared the relative thickness of layers across the V1 surface of a given animal or across animals. Nevertheless, it is clear from the literature that there is considerable similarity across animals. Accordingly, we cannot know what the source of variability in overall cortical thickness in our samples is, but we do see considerable consistency in the relative thickness of the layers we infer from our measures. We illustrate the differences that we see across penetrations and consider likely causes, such as the extent to which the coverslip pressing down on the cortex might differentially compress the cortex at different locations within the chamber.

The angle deviation of probe from surface will not change the relative thickness of layers, and the rigid linear probe is unlikely to bend in the cortex.

*Line 433. The term "Coherence" is used, clarify is this is you Cf from Figure 6. The text states, "marked decrease at the bottom of layer 6". Please clarify this, I do not see that in Figure 6.*

Text has been adjusted.

*In Figure 6, the locations of the lines between L1 and 2 do not seem to be consistent with respect to the subtle changes in light blue shading, across all three examples, yet the text*



*on line 436 states that there is a clear transition.*

We feel that the language used accurately reflects what is shown in the figure. While the transition is not sharp, it is clear that there is a transition. This transition is not used to define this laminar border. We have edited the text to clarify that the L1/2 border is better defined based on the change in AP power which shows a sharp transition (Figure 7).

*The text states that the boundary is also "always clear" from metrics... and sites Figure 5, but I do not see that this boundary is clear for all three examples in Figure 5.*

Text has been adjusted.

*Line 438. The text states that "it is not unusual for unit density to fall to zero below the L1/2 border (Figure 8E)", but surprisingly, the line in Figure 8 E does not even cover the indicated boundary between L1 and L2.*

At this point, the number of statements in the text that do not clearly and precisely correlate to the data in the figures is worrisome, and I think you could lose the confidence of readers at this point.

We do not see any inconstancy between what is stated in our text and what is noted by the reviewer. The termination of the blue line corresponds to the location where no units are detected. This is the location where "unit density falls to zero". In this example, no units resolved through spike sorting until ~100mm beneath the L1/L2 boundary, which is exactly zero unit density (Figure 8E). That there are electrical signals in this region is clear from the AP power change (Figure 8C) which also shows the location of the L1/L2 border.

*Line 448. Text states that the 6A/B border is defined by a sharp boundary in AP power, but Figure 8A "AP power spectrum" does not show a sharp change at the A/B line. There is a peak in this metric in the middle to upper middle of 6A, but nothing so sharp to define a boundary between distinct layers, at least for penetration A2.*

Text has been adjusted.

*In Figure 8, the layer labels are not clear, whereas they are reasonably clear in the other figures.*

This is a technical problem regarding vector graphics that were not properly converted in PDF generation. We will upload each high-quality vector graphics when we finalize the version of record.

*The text emphasizes differences in L4B and L4C with respect to average power and coherence, but the transition seems a bit gradual from layer 3B to 4C in some examples in Figure 6. And in Figure 5, A3, there doesn't appear to be any particular transition along the line between 4B and 4C.*

In this guide section, we pointed out early that some metrics are good for some boundaries and variation exists between penetrations. We've expanded text emphasizing the importance of timing differences in DP/P for differentiating sublayers in L4. Lastly, in case of several unresolvable boundaries given all the metrics, the prior knowledge of relative thickness should be used.

*Line 466 provides prescriptions in absolute linear distances, but this is unwise given that cortex may be crossed at oblique angles by electrodes, particularly for parts of V1 that*

are not on the surface of the brain. Other parts of the text have emphasized relative measurements.

Text has been changed using relative measurements.

*Line 507. The text says 9C and 4A are a good match, but the match does not look that good (4A has substantial dips at 0.5 and 0.75, and substantial peaks), and there is no quantification of fit. The error bars on 9C do not help show the variability across penetrations, they appear to be SEM, which shows that error bars get smaller as you average more data. It would seem more important to understand what is the variance in the density from one penetration to the next compared to the variance in density across layers.*

We have replaced “good match” with “roughly corresponds”. We note that we do not use unit density as a metric for identification of laminar borders and instead show that the expected locations of layers with higher neuronal density correspond to the locations where there are similar changes in unit density. It should be noted that Figure 9C is an average across many penetrations so should not be expected to show transitions that are as sharp in individual penetrations. Because of the figure permission issue, we have removed Figure 4A, and changed the text accordingly.

*Figure 9C-F show a lot of variability in the individual curves (dim gray lines) compared to the overall average. Does this show that these metrics are not reliable indicators at the level of single penetration, but show some trends across larger averages?*

In the beginning of the guide, we emphasized that all metrics should be combined for individual penetration, because some metrics are only reliable for delineating certain layer boundaries and the quality of data for the various measures varies between penetrations. The penetration average serves the same purpose explained in the previous question as an indicator that our layer delineation was not far off.

*The discussion mentions improvements in layer identification made here. Did this work check the assignments for these penetration against assignments made based on some form of ground truth? Previous methods would advance electrodes steadily, and make lesions, and carry out histology. Is there any way to tell how this method would compare to that?*

Even electrolytic lesions do not necessarily reveal ground truth and can be quite misleading. And their resolution is limited by lesion size. Lesions are typically variable in size, asymmetric and have variable shape and position relative to the location of the electrode tip, likely affected by the quality and location of electrical grounding and variations in current flow due to locations of blood vessels. A review of the published literature with electrode lesions shows that electrophysiological transitions are likely a far more accurate indicator of recording locations than post-mortem histology from electrolytic lesions. It is extremely rare for the locations of lesions to be precisely aligned to expected laminar transitions. See for example Chatterjee et al (Nature 2004). Also see several manuscripts from the Shapley lab. The lone rare exception of which we are aware is Blasdel and Fitzpatrick 1984 in which consistently small and round lesions were produced and even these would be too large (~100 microns) to accurately identify layers if it were not for the fact that the electrode penetrations were very long and tangential to the cortical layers.

### Reviewer #3 (Recommendations For The Authors):

- The authors say (lines 360-362) that "Assuming spikes of a neuron spread to at least two adjacent recording channels, then the coherence between the two channels would be directly proportional to number of spikes, independent of spike amplitude." Has this been demonstrated? Very large amplitude spikes should show up on more channels than small amplitude spikes. Do waveform amplitudes and unit densities from the spike waveform analyses show consistent relationships to the power and/or coherence distributions over depth across penetrations?

This part of the manuscript is providing a theoretical rational for what might be expected to affect the measures that we have derived. That is why we begin by stating that we are making an assumption. The answers to the reviewer's questions are not known and have not been demonstrated. By beginning with this theoretical preface, we can point to cases where the data match these expectations as well as other cases where the data differ from the theoretical expectations.

Coherence, by definition, is a normalized metric that is insensitive to amplitude. Spike amplitude mainly depends on how close the signal source is to electrode, and spike spread mainly depends on cell body size and shape given the same distance to electrode. Therefore, a very large spike amplitude could stem from a very close small cell to electrode, but would result in a small spike spread, especially axonal spikes (Figure 4B, red spike). Spike amplitudes on average are higher in L4C which matches the expectation that higher cell density would result, on average, closer cell body to electrode (Figure S4A). Nonetheless, the high-density small cell bodies in L4C result in a small spike spread (Figure 9D).

- I suggest clarifying what is defined as the baseline window for the  $\Delta P/P$  measure - is it the entire 10-150 ms response window used for the power spectrum analysis?

Text adjustments are made in the Methods where the time windows are defined at the beginning of the CSD section. Only temporal change metrics ( $\Delta CSD$  and  $\Delta P/P$ ) use the baseline window ([-40, 10]ms). The other two spectrum metrics (Power and Coherence) use the response window ([10, 150]ms).

- Firing rate differs by cell type and, on average, differs by layer in V1. Many layer 2/3 neurons, for example, have low maximum firing rates when driven with optimized achromatic grating stimuli. To the extent that the generative models explaining the sources of power and coherence signals rely on the assumption that firing rates are matched across cortical depth, these models may be inaccurate. This assumption is declared only subtly, and late in the paper, but it is relevant to earlier claims.

Text adjustments are made to explicitly describe the possibility that uneven depth profile of firing rate could counteract the depth profile of neuron density, resulting distorted or even a flat depth profile of power/coherence that deviates far from the depth profile of neuron density. In a newly added Figure S4, we first show the average firing rate profile during a set of stimuli (uniform color, static/drift, achromatic/chromatic gratings), then specifically the PSTHs of the same stimuli shown in this study. It can be seen that layers receiving direct LGN inputs tend to fire at a higher rate (L4C, L6A). Firing rates in the PSTHs either roughly match across layers or are much higher in the densely packed layers. Therefore, the depth profile of firing rate contributes to rather than counteracting that of neuron density, enhancing the utility of the power/coherence profile for identification of correct layer boundaries.

- Given the acute preparation used for recordings, I wonder whether tissue is available for histological evaluation. Although the layers identified are generally appropriate in

*relative size, it would be particularly compelling if the authors could demonstrate that the fraction of the cortical thickness occupied by each layer corresponded to the proportion occupied by that layer along the probe trajectory in histological sections. This would lend strength to the claim that these analyses can be used to identify layers in the absence of histology. Furthermore, variations in apparent cortical thickness could arise from different degrees of deviation from surface normal approach angles, which might be apparent by evaluation of histological material. I would add that variation in thickness on the scale shown in Fig. S4 is more likely to have an explanation of this kind.*

To serve other purposes unrelated to this study (identification of CO blobs), we cut the postmortem tissue in horizontal slices, so the histological comparison suggested cannot be made. The cortical thickness measured in this study had been affected not only by the angle deviation from the surface normal but also the swelling and compression of cortex. Nevertheless, evaluating the absolute thickness of cortex is not the main purpose of this study.

*Text and figure suggestions:*

*- Fig 1A has been modified from Balaram & Kaas (2014) to revert to the Brodmann nomenclature scheme they argue against using in that paper; I wonder if they would object to this modification without explanation. Related, in the main text the authors initially refer to layers using Brodmann's labels with a secondary scheme (Hassler's) in parentheses and later drop the parenthetical labels; these conventions are not described or explained. Readers less familiar with the multiple nomenclature schemes for monkey V1 layers might be confused by the multiple labels without context, and could benefit from a brief description of the convention the authors have adopted.*

Throughout our article, we only used Brodmann's naming convention because it has historically been adopted for old world monkey which we use in our study, whereas Hassler's naming convention is more commonly used for new world monkey. Different naming conventions do not change our result, and it is out of scope for our study to discuss which nomenclature is more appropriate.

*- References to "dominate eye" throughout the text and figure legends should be replaced with "dominant eye."*

It has been changed throughout the article.

*- It is a bit odd to duplicate the same example in Fig. 2C and 2E. Perhaps a unique example would be a better use of the space.*

Here we first demonstrate the filtering effect, then compare profiles across different penetrations. The same example bridges the transition allowing side-by-side comparison.

*- The legend for Fig. 3 might be clearer if it simply listed the stimulus transitions for each column left to right, i.e. "black to white (non-dominant eye), white to black (non-dominant eye), black to white (dominant eye), ..."*

We feel that the icons are helpful. Here we want to show the stimulus colors directly to readers.

*- The misalignment between Fig. 4A vs. 4B-F, combined with the very small font size of the layer labels in Fig. 4B-F, make the visual comparison difficult. In Figs. 7 and 8, layer labels (and most labels in general) are much too small and/or low resolution to read easily.*

| *Overall, I would recommend increasing font size of labels in figures throughout the paper.*

The reused figure in Figure 4A is removed due to permission issue. Font sizes are adjusted.

| *- Line 591 "using of high-density probes" should be "using high-density probes"*

Text has been changed accordingly

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